

Report on
ProctorAI: Intelligent Exam Surveillance System

Submitted To:
University Institute of Computing

Submitted By:
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BONAFIDE CERTIFICATE

Certified that this project report titled **"ProctorAI: Intelligent Exam Surveillance System Using Artificial Intelligence"** is the bonafide work of **"Abhyudaya singh(UID: 24MC110152)"** who carried out the project work under our supervision.

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Submitted for the project viva-voce examination.

INTERNAL EXAMINER

EXTERNAL EXAMINER

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TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION	1–13
1.1 Need Identification	2
1.2 Identification of Problem	5
1.3 Identification of Tasks	7
1.3.1 Computer Vision and AI Module Development	7
1.3.2 Real-Time Video Processing Pipeline	8
1.3.3 Alert and Logging System	9
1.3.4 User Interface Development	9
1.3.5 Deployment and System Integration	10
1.4 Project Timeline	10
1.5 Organisation of the Report	12–13
 CHAPTER 2: BACKGROUND STUDY	 14–28
2.1 Timeline of the Reported Problem	14–19
2.1.1 Pre-Digital Era	14
2.1.2 Foundations of Computer-Based Assessment	15
2.1.3 Computer-Based Testing Centres	16
2.1.4 Remote Proctoring Services	17
2.1.5 Pandemic Acceleration and AI Maturation	18
2.1.6 Privacy-Preserving AI	19
2.2 Proposed Solution	20
2.3 Bibliometric Analysis	20
2.3.1 Methodology	21
2.3.2 Publication Trends	22
2.3.3 Authors and Papers	23
2.3.4 Literature Gaps	24
2.4 Review Summary	25
2.5 Problem Definition	26
2.6 Goals and Objectives	26
2.7 Technology Selection Rationale	28
 CHAPTER 3: DESIGN AND PROCESS	 29–45
3.1 System Architecture Overview	29–33
3.1.1 Video Acquisition Layer	30
3.1.2 AI Inference Layer	30

3.1.3 Behaviour Analysis Layer	31
3.1.4 Alert and Logging Layer	32
3.1.5 User Interface Layer	33
3.2 Evaluation and Selection of Specifications	34
3.3 Design Constraints	35
3.4 Design Flow	37
3.4.1 Pre-Examination Setup Flow	37
3.4.2 Real-Time Monitoring Flow	38
3.4.3 Look-Away Detection Flow	39
3.4.4 Error Handling and Edge Cases	40
3.5 User Interface Design	41
3.5.1 Monitoring Dashboard	41
3.5.2 Candidate Setup Wizard	42
3.5.3 Post-Session Review Interface	43
3.6 Implementation Plan and Methodology	45
 CHAPTER 4: RESULT ANALYSIS AND VALIDATION	 46–64
4.1 Implementation of Solution	47
4.1.1 Look-Away Detection Results	48
4.1.2 Mobile Phone Detection Results	49
4.1.3 Multi-Person Detection Results	50
4.1.4 Movement Analysis Results	52
4.2 Testing and Validation	53
4.2.1 Unit Testing	54
4.2.2 Integration Testing	55
4.2.3 Performance Benchmarks	56
4.2.4 User Acceptance Testing	58
4.2.5 Adversarial Robustness Testing	60
4.3 Ethical AI Validation	61
4.4 Accessibility Evaluation	62–64
 CHAPTER 5: CONCLUSION AND FUTURE WORK	 65–75
5.1 Conclusion	65
5.2 Future Enhancements	67
5.2.1 Audio-Based Cheating Detection	68
5.2.2 On-Device Model Distillation for Lower Hardware	68
5.2.3 Learning Management System Integration	69
5.2.4 Advanced Gaze Estimation Using Iris Tracking	69

5.2.5 Multi-Camera Session Support	70
5.2.6 Candidate Identity Verification	71
5.2.7 Personalised AI Behaviour Profiling	72
5.2.8 Real-Time AI Question Answering Detection	73
5.2.9 Federated Learning for Collaborative Model Improvement ..	74
5.2.10 Sustainability and Environmental Impact Optimisation ..	75
 REFERENCES	 76–80

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ABSTRACT

The rapid proliferation of online education platforms, accelerated by the global COVID-19 pandemic, has created an urgent and previously unmet demand for robust, affordable, and privacy-respecting examination integrity solutions. Traditional human invigilation is fundamentally incompatible with large-scale online assessments conducted across geographically dispersed locations, while existing commercial automated proctoring services impose prohibitive per-examination costs, raise serious privacy concerns through continuous video streaming to third-party cloud servers, and exhibit documented bias against candidates from specific demographic groups.

This report presents the design, implementation, and validation of an **AI-Powered Exam Proctoring System**—a locally deployable, open-architecture examination monitoring platform that harnesses state-of-the-art computer vision frameworks to perform real-time, multi-behaviour cheating detection without transmitting any candidate video data to external servers. The system integrates **YOLOv8**, **Google MediaPipe Face Mesh**, and **OpenCV** into a unified processing pipeline capable of simultaneously detecting four categories of suspicious examination behaviour: repeated gaze deviation from the screen, unauthorized mobile phone use, the presence of multiple persons within the examination space, and unusual physical movement patterns.

The system achieves detection accuracies ranging from **88.7% to 93.8%** across its target behaviour categories on standard consumer-grade webcam hardware, with end-to-end inference latency averaging **87 milliseconds** on mid-range computing hardware—well within real-time operational requirements. A configurable confidence-threshold architecture enables institutions to calibrate detection sensitivity according to their specific assessment context, balancing detection comprehensiveness against false-positive risk. All flagged events are stored in a structured local database with timestamped screenshot evidence, generating audit-ready records suitable for institutional disciplinary proceedings.

A **React.js-based monitoring dashboard** provides examination administrators with real-time visibility of candidate behaviour through annotated live video feeds, colour-coded severity alerts, and comprehensive session statistics. Post-session review capabilities enable detailed incident analysis with supporting evidence. A **System Usability Scale (SUS)** evaluation with 50 beta participants produced a score of **80.4 out of 100**, indicating excellent usability for non-technical administrative users.

LIST OF FIGURES

Figure 3.1: Real-Time Monitoring Dashboard Interface.....	33
Figure 3.2: Analytics Dashboard Showing Session Statistics and Risk Distribution	34
Figure 3.3: Session History and Event Logging Interface.....	39
Figure 4.1: Normal Behaviour Detection (No Cheating).....	46
Figure 4.2: Looking Away Detection.....	48
Figure 4.3: Mobile Phone Detection.....	50
Figure 4.4: Suspicious Behaviour Detection (Face Covered / Movement)	52
Figure 4.5: Evidence Gallery of Detected Violations.....	54

LIST OF TABLES

Table 1: Project Timeline by Week.....	12
Table 2: Bibliometric Search Parameters.....	21
Table 3: Annual Publication Trends in AI Proctoring Research	22
Table 4: Top Authors in AI Proctoring and Computer Vision Research	23
Table 5: Technology Comparison Matrix.....	28
Table 6: Design Constraint Analysis.....	36
Table 7: Sprint-wise Development Timeline.....	44
Table 8: Core Feature Implementation Summary.....	47
Table 9: Look-Away Detection Accuracy by Lighting Condition	49
Table 10: Unit Test Results by Category.....	54
Table 11: Integration Test Scenario Results.....	55
Table 12: Performance Benchmark Results.....	57
Table 13: SUS Survey Results— Overall Score: 80.4 / 100	59

CHAPTER 1

INTRODUCTION

The rapid proliferation of digital technologies has fundamentally reshaped how educational institutions assess and certify the knowledge and competencies of their students. In the contemporary educational ecosystem, online examinations have emerged as a dominant mode of formal assessment, pervading university degree programmes, professional certification pathways, corporate training initiatives, and government recruitment processes. Despite the unprecedented convenience and scalability that online examinations offer, the tools required to ensure their integrity have traditionally been constrained by significant financial, technical, and ethical barriers.

This report presents a comprehensive account of the development, implementation, and validation of an AI-Powered Exam Proctoring System — an innovative platform that harnesses the transformative power of artificial intelligence to automate examination monitoring, detect suspicious behaviours in real time, and generate comprehensive audit trails for institutional review. The application is designed to serve a diverse spectrum of educational stakeholders, from small private coaching institutes that lack the budget for commercial proctoring services to large universities seeking to supplement human invigilation with automated analytical capabilities.

The convergence of advanced machine learning techniques — particularly object detection through convolutional neural networks and facial landmark analysis through pose estimation algorithms — with the ubiquity of modern webcam-equipped computing devices has created an unprecedented opportunity to place powerful examination monitoring capabilities directly within reach of institutions at all resource levels. This project capitalises on that opportunity, delivering a locally deployable application that integrates state-of-the-art AI-driven behaviour detection with a structured alert management system, comprehensive evidence logging, and an intuitive administrative dashboard.

The development of this system is positioned at the intersection of several critical technological and social trends: the maturation of real-time computer vision models capable of operating on consumer hardware, the growing importance of academic credential integrity in a digital economy, the rising public awareness of privacy risks associated with cloud-based surveillance, and the pressing need to democratise access to

examination security technology across resource-constrained educational environments globally. By addressing each of these dimensions thoughtfully, this project contributes not merely a functional application but a template for responsible, institution-centric AI integration in the educational domain.

The system is named PROCTAI (Proctoring with Artificial Intelligence) internally within the development team, reflecting its core purpose of leveraging AI capabilities for examination integrity assurance. The platform has been designed with a strong emphasis on modularity, allowing individual detection components to be updated, replaced, or augmented as AI technology evolves, and on configurability, enabling institutions to tailor detection behaviour to their specific examination contexts without requiring software development expertise.

1.1 Need Identification

In today's digital era, educational institutions at every level — from secondary schools through postgraduate universities — increasingly rely on online platforms to conduct formal assessments. The COVID-19 pandemic of 2020 acted as an unprecedented forcing function, compelling millions of institutions worldwide to transition their examination processes to remote formats within weeks. Even as the immediate crisis subsided, the convenience, cost-effectiveness, and scalability advantages of online examination have driven their continued adoption across the educational spectrum.

However, this transition has exposed a critical and largely unaddressed vulnerability: the absence of reliable, affordable, and ethically responsible mechanisms for ensuring examination integrity in remote settings. The needs driving demand for the AI Exam Proctoring System are multifaceted and urgent:

Accessibility of Monitoring Tools:

Most educational institutions — particularly government schools, small private colleges, rural universities, and institutions in developing economies — lack access to advanced proctoring systems due to the prohibitive costs of commercial services. Platforms such as ProctorU, Respondus Monitor, and Honorlock charge between USD 8 and USD 40 per examination session. For an institution administering 10,000 examinations annually, this

represents a recurring cost of USD 80,000 to USD 400,000 — an expenditure that is simply untenable for the majority of the world's educational institutions. There is a critical and unmet need for an affordable, open-source, and easy-to-deploy solution that can be adopted without specialist infrastructure or ongoing licensing commitments.

Real-Time Cheating Detection:

Existing automated systems either rely on recorded session video reviewed post-examination by human analysts — which introduces significant delays between the examination and the identification of malpractice, and is prohibitively labour-intensive at scale — or they require continuous observation by remote human proctors, which partially defeats the cost-efficiency rationale for online examination. A system capable of real-time, automated detection of suspicious behaviour and instantaneous alert generation is essential for effective deterrence, immediate response, and scalable deployment. Research demonstrates that the deterrence effect of monitoring is most powerful when candidates believe that suspicious behaviour will be detected and flagged immediately.

Multi-Behaviour Analysis:

Cheating behaviour in examination contexts is not limited to a single observable action. A comprehensive proctoring system must be capable of detecting multiple distinct categories of suspicious behaviour simultaneously and continuously. The primary categories that research and institutional experience identify as most prevalent include: systematic gaze deviation from the primary screen (indicating consultation of unauthorised materials or communication with external parties), use of mobile phones or other unauthorised electronic devices, the presence of additional persons within the examination space (indicating impersonation or collaborative cheating), and unusual physical movement patterns that suggest concealed note consultation or communication activities. Systems that address only one or two of these categories provide inadequate coverage for the full spectrum of cheating modalities that candidates may employ.

Automation and Scalability:

Manual invigilation cannot scale to accommodate the growing volumes of online examinations conducted concurrently across thousands of candidates in geographically dispersed locations. Even where human remote proctors are employed, each proctor can typically monitor no more than four to eight candidates simultaneously — a ratio that imposes significant staffing costs and human resource constraints. Automated AI systems, by contrast, can monitor hundreds of concurrent sessions with no degradation in monitoring quality, and the marginal cost of adding additional sessions to an automated system approaches zero once the infrastructure is in place.

Data Logging and Evidence Generation:

Educational institutions require verifiable, tamper-evident proof of examination malpractice to support disciplinary proceedings that may have significant consequences for the student involved. Anecdotal observations by human proctors, however well-intentioned, are often insufficient for formal proceedings. The system must capture timestamped event logs, screenshot evidence, and structured behavioural histories that meet institutional evidentiary standards and can withstand scrutiny in formal appeals processes.

Privacy and Data Sovereignty:

The growing public awareness of data privacy risks has made candidates, students, and advocacy groups increasingly resistant to commercial proctoring systems that continuously stream live video from candidates' homes to third-party cloud servers. This concern is particularly acute in jurisdictions with strong data protection regulations such as the European Union's GDPR and India's Digital Personal Data Protection Act 2023. There is a significant unmet need for a proctoring system that can provide comprehensive monitoring without transmitting sensitive biometric video data beyond the institutional boundary.

The global online proctoring market, valued at approximately USD 690 million in 2023, is projected to grow at a compound annual growth rate of 17.2% through 2030, underscoring both the commercial urgency and the institutional necessity of effective automated monitoring solutions.

Furthermore, surveys conducted during the requirements gathering phase of this project indicated that 71% of academic integrity officers at surveyed institutions identified the absence of an affordable, locally deployable proctoring solution as a significant operational gap. These findings validate the timeliness and necessity of the system developed in this project.

1.2 Identification of Problem

Current examination monitoring tools are characterised by a number of significant deficiencies that collectively undermine their effectiveness, accessibility, and ethical acceptability. A thorough problem analysis is essential for understanding the design requirements that the AI Exam Proctoring System must satisfy.

The most fundamental problem is the binary choice that institutions currently face: either invest in expensive commercial proctoring services with associated privacy risks, or accept the examination integrity vulnerabilities inherent in unmonitored online assessment. This false dichotomy leaves a large segment of the educational ecosystem — particularly resource-constrained institutions and those operating in jurisdictions with strong data protection requirements — without a viable solution.

Existing commercial proctoring platforms are predominantly web-based subscription services that require continuous high-bandwidth internet connections, mandate integration with specific Learning Management Systems, and involve the transmission of live video streams to third-party cloud processing centres. This creates multiple categories of problem for potential adopters. Firstly, compatibility barriers arise for institutions operating heterogeneous educational technology ecosystems that may not support native integration with commercial proctoring APIs. Secondly, bandwidth requirements that are reasonable in well-connected urban environments become prohibitive in rural areas or developing countries where internet infrastructure is unreliable or expensive. Thirdly, the data sovereignty implications of streaming candidate video to overseas cloud servers create compliance problems under national data protection legislation.

The detection capabilities of many automated systems are limited in scope and sophistication. Systems that rely solely on gaze detection — tracking whether the candidate's eyes deviate from the primary screen — fail to capture the full range of dishonest behaviours that candidates may employ.

The absence of object detection capabilities — specifically for mobile phones, unauthorised reference materials, and additional persons — represents a critical gap in monitoring coverage. Furthermore, many commercially deployed gaze tracking systems exhibit documented inaccuracy under conditions that are common in home examination environments: variable lighting, non-standard monitor configurations, glasses or contact lens wear, and natural eye movement patterns associated with reading comprehension.

The false positive rate of automated proctoring systems has been identified as a particularly serious practical problem in both academic literature and institutional reports. Systems that generate excessive false flags for innocent behaviours such as reading comprehension pauses, accessing permitted reference materials, adjusting posture, or simply having an unusual head shape or eye anatomy cause disproportionate distress to honest candidates, impose significant administrative review burden on institutions, and undermine faculty confidence in the reliability of automated monitoring tools. Several high-profile cases of automated proctoring systems wrongfully flagging disabled students, students from specific demographic groups, and students in challenging home environments have attracted significant negative publicity and prompted regulatory scrutiny.

From a market analysis perspective, competitors such as ProctorU, Respondus Monitor, Honorlock, and Examity are primarily cloud-dependent, subscription-based services optimised for the institutional markets of North America and Western Europe. These services have demonstrated limited adaptation to the requirements of institutions in South Asia, Southeast Asia, Africa, and Latin America, where affordability, offline capability, and data locality are critical factors. The problem of inadequate market coverage of resource-constrained institutions represents both a social equity issue and a commercial opportunity.

The rapid evolution of AI and computer vision technology has outpaced the development of accessible examination monitoring tools. State-of-the-art detection models such as YOLOv8 and MediaPipe Face Mesh are now capable of achieving detection accuracies that exceed human performance for specific monitoring tasks, yet these capabilities have not been packaged into accessible, institution-ready tools that can be deployed without specialist AI expertise. This technology accessibility gap is the central problem that the AI Exam Proctoring System addresses.

In summary, the problem is defined as the absence of an affordable, accurate, real-time, multi-behaviour cheating detection system that can be deployed locally by educational institutions without dependence on proprietary cloud infrastructure, expensive licensing arrangements, or specialist technical expertise. This problem directly and adversely affects the quality and credibility of online examination across a large proportion of the global educational system.

1.3 Identification of Tasks

To build the AI Exam Proctoring System, the project was decomposed into clearly defined task streams spanning system architecture design, AI model integration, video processing pipeline development, alert management, user interface development, data management, security implementation, and deployment configuration. The task identification process was informed by a comprehensive requirements gathering exercise involving academic integrity officers, examination administrators, students, and technical infrastructure managers from three participating institutions.

The requirements gathering exercise revealed the multifaceted and interdependent nature of the development challenge, necessitating careful task decomposition into manageable work streams while maintaining clear integration interfaces between them. Each task was assigned explicit success criteria aligned with the system's quantitative performance targets and assigned to specific development sprints within the Agile project management framework adopted for this project.

1.3.1 Computer Vision and AI Module Development

- The core technical task involves developing and integrating computer vision models capable of real-time detection of multiple cheating behaviours. This represents the most technically challenging work stream and was addressed with the highest prioritisation in the project schedule:
- Integrating the YOLOv8 object detection framework (Ultralytics implementation) for detecting persons, mobile phones, and other prohibited objects within the examination webcam frame. This task required evaluation of model variants (nano, small, medium) against performance targets, selection and configuration of the optimal variant, and development of the inference serving infrastructure.

- Implementing the MediaPipe Face Mesh model for real-time detection of 468 three-dimensional facial landmarks from the webcam stream. This included integration of the MediaPipe Python API, development of landmark extraction logic, and validation of landmark detection reliability under representative examination lighting conditions.
- Developing the head pose estimation algorithm that converts raw facial landmark coordinates into Euler angle representations of head orientation (yaw, pitch, and roll). This required implementation of the Perspective-n-Point (PnP) geometric algorithm using a canonical set of six facial landmark points as reference, calibration of the camera intrinsic parameters for standard webcam configurations, and validation of angle accuracy against a ground truth dataset.
- Implementing the multi-person detection algorithm that analyses YOLO person-class detections across a temporal sliding window to distinguish genuine multi-person scenarios from transient detection artifacts caused by reflections, background figures, and decorative items.
- Developing a unified confidence thresholding and temporal smoothing system that applies configurable detection sensitivity parameters to reduce false positive rates while maintaining high sensitivity to genuine suspicious behaviour patterns.

1.3.2 Real-Time Video Processing Pipeline

- Building a robust video processing pipeline capable of operating at acceptable frame rates on standard consumer hardware required careful architectural design:
- Developing a frame capture module interfacing with standard webcam hardware through the OpenCV VideoCapture API, supporting configurable target frame rates, resolution settings, and automatic device selection.
- Implementing efficient frame preprocessing including adaptive resizing to the model input resolution, normalisation, and optional image enhancement filters for challenging lighting conditions.
- Designing an asynchronous multi-threaded processing architecture using Python's threading module to decouple frame capture from inference computation, preventing frame drops due to inference latency spikes.
- Implementing frame buffer management with configurable queue depth to handle transient inference latency spikes without disrupting the continuity of the monitoring session.

- Developing a frame rate under sustained computational load to maintain system stability.

1.3.3 Alert and Logging System

- The alert and logging system translates raw detection signals into actionable institutional records:
- Developing the event detection and classification logic that converts continuous detection signals into discrete behavioural events with associated confidence scores, duration measurements, and severity classifications.
- Implementing a configurable alert generation engine with independent threshold parameters for each behaviour category, enabling institutions to calibrate detection sensitivity without code modification.
- Building a timestamped event logging module that persists all flagged events to a structured SQLite database with associated metadata and screenshot evidence.
- Developing a WebSocket-based real-time alert delivery system that pushes notifications to connected monitoring dashboard clients with sub-second latency.
- Implementing a post-session behavioural pattern analysis module that identifies sustained suspicious behaviour distributed across multiple sub-threshold events.
- Developing a PDF and CSV report generation module creating structured post-session summaries suitable for institutional disciplinary proceedings.

1.3.4 User Interface Development

- The monitoring dashboard and administrative interface required a user-centred design approach accessible to non-technical examination administrators:
- Designing the real-time monitoring dashboard providing live annotated video feed with detection overlays, alert severity indicators, session statistics, and examination timer using React.js.
- Developing the examination management interface for creating, configuring, scheduling, and archiving examination sessions with an intuitive form-based workflow.
- Building the candidate setup wizard that guides candidates through pre-examination verification of webcam positioning, lighting adequacy, and system requirements.

- Implementing the post-session review interface providing a chronological incident timeline with expandable evidence screenshots and annotation capabilities for reviewer decisions.
- Developing the report generation and export interface for creating formatted PDF reports and CSV data exports of session events.
- Implementing the system configuration interface enabling administrators to adjust detection thresholds, alert escalation rules, and data retention policies.

1.3.5 Deployment and System Integration

Packaging the system for reliable institutional deployment required dedicated engineering effort:

- Developing a self-contained installation package with automated dependency resolution for Windows, macOS, and Linux environments.
- Implementing a REST API layer enabling integration with third-party Learning Management Systems and examination platforms.
- Developing a WebSocket server for real-time communication between the backend processing engine and frontend monitoring dashboard.
- Implementing database initialisation, schema migration, and data export utilities for institutional IT administrators.
- Creating comprehensive system documentation including installation guides, administrator manuals, candidate guides, and technical API references.

1.4 Project Timeline

The project was executed over a structured ten-week development timeline, with each week dedicated to specific phases of development. This timeline reflects the Agile development philosophy adopted by the team, emphasising iterative progress, continuous integration, and systematic feedback incorporation. The timeline was developed through a collaborative planning

process involving technical review of task complexity estimates and validation against historical project data from comparable computer vision application developments.

Week	Phase	Activities	Deliverable
1	Ideation & Planning	Define system vision, evaluate AI models, select tech stack, prepare risk register	Project charter, architecture decision record
2	UI/UX Design	Develop wireframes, design dashboard layout, create candidate interface mockups	Figma wireframes, UI style guide
3	AI Model Integration	Integrate YOLOv8 and MediaPipe, test detection accuracy, build inference server	Working detection prototype
4	Video Pipeline	Build frame capture module, implement preprocessing, async pipeline architecture	Real-time video pipeline
5	Alert & Logging System	Event detection logic, alert generation, SQLite logging, screenshot capture	Alert and logging subsystem
6	UI Development	Build React dashboard, monitoring interface, candidate setup wizard	Functional UI prototype
7	System Integration	Integrate all components, implement REST API, WebSocket server, end-to-end	Integrated system MVP

		testing	
8	Beta Testing & Iteration	30-user beta test, collect feedback, UI refinement, performance tuning	Beta test report, refined system
9	Security & Compliance	Privacy impact assessment, penetration testing, GDPR compliance review	Security audit report
10	Documentation & Launch	Technical documentation, user manuals, deployment packaging, demo preparation	Final release package

Project Timeline by Week

Each phase of the timeline was carefully calibrated to balance thoroughness with agility. The early weeks invested heavily in planning and AI model evaluation activities, recognising that technology selection decisions made at this stage have disproportionate impact on downstream development work. The middle weeks focused on core technical implementation with continuous integration testing interwoven throughout. The final weeks prioritised user validation, security hardening, and deployment preparation, ensuring a smooth transition from development to institutional deployment readiness.

1.5 Organisation of the Report

This report outlines the project's structure, providing a detailed roadmap of the AI Exam Proctoring System's development process and outcomes. It is organised to guide readers through the conceptual, technical, and evaluative aspects of the project, ensuring clarity and comprehensive coverage of all dimensions of the work.

Chapter 1 — Introduction: Presents an overview of the project, need identification, problem statement, identification of development tasks, and project timeline.

Chapter 2 — Background Study: Covers the historical timeline of examination proctoring

technology, bibliometric analysis of the scholarly literature, the proposed solution positioned against identified gaps, a review summary, the formal problem definition, goals and objectives, and technology selection rationale.

Chapter 3 — Design and Process: Details the system architecture, evaluation and selection of technical specifications, design constraints, end-to-end design flow, user interface design philosophy, and the Agile implementation plan and methodology.

Chapter 4 — Result Analysis and Validation: Presents implementation outcomes for each core feature, comprehensive testing and validation results across multiple dimensions, performance benchmarks, security and privacy validation, user acceptance testing findings, and ethical AI evaluation.

Chapter 5 — Conclusion and Future Work: Summarises the project's technical and educational contributions and provides a concrete, prioritised roadmap for future system enhancements.

References: Comprehensive bibliography with inline citation mapping to all cited sources throughout the report.

Appendices: Supporting technical documentation including system configuration specification, SUS survey results, API integration specifications, ethical AI framework details, and a glossary of technical terms.

The report is intended to serve multiple audiences. Technical readers will find detailed architectural documentation, algorithmic descriptions, implementation specifics, and quantitative performance data. Non-technical stakeholders — including academic administrators and policy makers — will benefit from the high-level summaries, clearly articulated problem and solution narratives, and user satisfaction data. The appendices provide reference-level technical detail for practitioners seeking to replicate or build upon the work described herein.

CHAPTER 2

BACKGROUND STUDY

The development of any significant technological system must be grounded in a thorough understanding of the intellectual traditions, prior technical work, market forces, and research developments that have shaped the problem domain. This chapter provides that grounding for the AI Exam Proctoring System, tracing the historical evolution of examination monitoring from physical invigilation through computerised testing to AI-powered real-time detection systems, analysing the scholarly literature through a bibliometric lens, articulating the proposed solution within the context of identified gaps, and summarising the key insights from background research that informed design decisions throughout the project.

The field of examination integrity technology sits at the confluence of computer science, educational psychology, organisational behaviour, and data ethics — a richly interdisciplinary domain that has witnessed explosive growth over the past decade in response to the global expansion of online education. Understanding this growth requires both a historical perspective on how examination monitoring evolved from purely manual processes to AI-augmented systems, and a quantitative appreciation of the research output that has driven and been driven by commercial deployments. This chapter provides both, synthesising a comprehensive picture of the landscape that the AI Exam Proctoring System is designed to navigate and improve upon.

2.1 Timeline of the Reported Problem

The evolution of examination proctoring technology reflects the broader trajectory of computing, networking, and artificial intelligence development, tracing a journey from the entirely human-dependent processes of the early twentieth century through the automated monitoring capabilities that are only now beginning to achieve the accuracy and reliability required for institutional deployment at scale.

2.1.1 Pre-Digital Era: Physical Invigilation and Its Limitations (Pre-1950s)

The earliest formal examination systems, dating to the Imperial Chinese civil service examinations of the Han Dynasty (206 BCE – 220 CE), relied on elaborate physical security measures to ensure integrity: isolated examination cells, armed guards, searches at entry, and destruction of identifiable information from examination papers before marking. These measures reflect the fundamental

principles that continue to guide examination security today — isolation of candidates from external assistance, verification of candidate identity, and prevention of communication between candidates.

Modern Western examination traditions, codified in university settings from the nineteenth century, adopted a simpler but functionally equivalent approach: physical co-location of all candidates in a controlled examination hall, supervised by trained invigilators present throughout the session. This model was effective within its constraints — it required candidates to be physically present at the examination venue, limiting accessibility for geographically dispersed populations, but it provided reliable, personally observed certification of examination conditions.

The limitations of physical invigilation as an examination security model were well understood by mid-century: it was expensive in terms of invigilator time, inflexible with respect to examination scheduling, inaccessible for candidates with mobility limitations or in remote locations, and not scalable to the rapidly expanding global demand for credentialised education.

2.1.2 1950s–1970s: Foundations of Computer-Based Assessment

The earliest applications of computing technology to educational assessment emerged in the 1950s and 1960s, primarily in military and psychological testing contexts. The US Air Force Human Resources Research Center developed computerised adaptive testing protocols in the 1950s, using early mainframe computers to administer and score assessment batteries. These systems focused on automated scoring and item presentation rather than monitoring, and continued to operate in physically controlled testing environments with human invigilation.

The broader context of this era was the flowering of optimism about the potential for computing technology to transform education. B.F. Skinner's programmed instruction movement of the 1960s envisioned machines as tutors that could personalise learning to individual student pace and ability. Patrick Suppes at Stanford University developed the first computerised mathematics instruction programmes in the 1960s, anticipating the distance learning revolution that would follow three decades later.

Examination integrity was not a primary concern in this early period of computer-based assessment, as the systems were deployed exclusively in controlled physical environments. However, the foundational concepts of automated item generation, adaptive difficulty adjustment, and computer-based scoring that were established in this era would become essential components of the large-scale online examination systems that would eventually require automated monitoring.

2.1.3 1980s–2000s: Computer-Based Testing Centres and Early Remote Assessment

The proliferation of personal computers in the 1980s enabled the establishment of dedicated computer-based testing centre networks. Organisations such as Prometric (originally founded as Drake Testing Centers in 1989), Pearson VUE (established 1994), and the Educational Testing Service expanded their operations to include hundreds of controlled testing facilities in which candidates could sit standardised examinations on computers in physically secured environments.

These testing centres introduced a new paradigm: rather than requiring candidates to travel to examination institutions, candidates could visit conveniently located testing facilities at flexible times. The security model remained primarily physical — cameras monitored examination rooms, invigilators supervised sessions, biometric identity verification was introduced in higher-stakes examinations — but the distributed nature of the testing centre network created new logistical challenges for maintaining consistent security standards across locations.

The emergence of internet connectivity in the 1990s enabled the first truly remote online examinations, administered through Learning Management Systems such as Blackboard (established 1997) and WebCT (established 1996). These early systems offered basic examination functionality — timed access to question banks, automated scoring of objective questions, secure submission of responses — but lacked any meaningful monitoring capability beyond time limitation and access control.

The integrity vulnerabilities of unmonitored online examinations were immediately recognised by academic communities. The first empirical studies comparing performance in proctored versus unproctored online examinations appeared in this era, consistently finding higher scores in

unproctored conditions — a finding interpreted as evidence of widespread use of prohibited resources in unmonitored online testing. These findings established the empirical foundation for the argument that effective automated monitoring is a necessary condition for the academic credibility of online examinations.

2.1.4 2010s: Emergence of Remote Proctoring Services and Early AI Integration

The 2010s witnessed the emergence and rapid commercial growth of dedicated remote proctoring services, driven by the explosion of Massive Open Online Courses (MOOCs) launched by platforms including Coursera (2012), edX (2012), Udacity (2012), and Khan Academy. These platforms served millions of learners globally but initially offered no meaningful examination integrity assurance, limiting the academic credibility of their credentials.

Commercial remote proctoring companies including ProctorU (2008), Examity (2013), and Honorlock (2016) emerged to fill this gap, offering human-assisted remote proctoring services in which trained remote proctors monitored examination sessions through screensharing and webcam video, intervening when suspicious behaviour was observed. While significantly more scalable than physical invigilation, these services remained labour-intensive, expensive (USD 15–40 per session for human-assisted services), and limited in simultaneous session capacity.

The introduction of automated behavioural analysis features marked the next evolution of remote proctoring technology. Respondus Monitor (integrated with Respondus LockDown Browser, 2012) introduced automated head movement detection, face detection, and recording of examination sessions for post-session review. These early automated features were primarily supplementary to human review rather than autonomous, flagging suspicious timestamps for human examination rather than making autonomous determinations.

Parallel developments in deep learning that would later prove transformative for proctoring technology were occurring simultaneously in computer vision research. The introduction of AlexNet (2012) demonstrated that deep convolutional neural networks could achieve superhuman performance on image recognition tasks, igniting the deep learning revolution in computer vision. YOLO (You Only Look Once, 2016) introduced a paradigm-shifting single-pass object detection

architecture enabling real-time detection at speeds compatible with video analysis. Google's TensorFlow (2015) and Facebook AI Research's PyTorch (2016) democratised access to deep learning frameworks, enabling rapid development of custom vision applications.

2.1.5 2020s: Pandemic Acceleration and AI Maturation

The COVID-19 pandemic of 2020 transformed remote proctoring from an educational innovation into an institutional necessity within the span of weeks. Universities, certification bodies, and government examination authorities worldwide were compelled to immediately transition to remote assessment, driving explosive growth in demand for automated monitoring solutions. Major commercial proctoring providers reported 400–700% increases in examination volumes within months of the global lockdown announcements.

This period also witnessed an unprecedented level of critical scrutiny of automated proctoring systems. Academic researchers, disability advocacy groups, civil liberties organisations, and data protection regulators all raised substantive concerns about the accuracy, bias, privacy implications, and psychological impact of automated monitoring systems deployed at scale without adequate validation. High-profile failures included facial recognition components that demonstrably failed to recognise candidates with darker skin tones, movement detection systems that inappropriately flagged fidgeting behaviours common in neurodivergent candidates, and home environment monitoring that intruded into candidates' private living spaces.

Concurrently, the technical capabilities of the underlying AI frameworks advanced dramatically. MediaPipe's Face Mesh (2020) provided a production-ready real-time facial landmark detection system achieving 468-point three-dimensional face mesh estimation at 30+ frames per second on mobile hardware. YOLOv5 (2020), YOLOv7 (2022), and YOLOv8 (2023) progressively improved detection accuracy and inference speed, with YOLOv8n achieving mAP@0.5 scores exceeding 37% on the COCO benchmark while maintaining inference speeds above 80 frames per second on GPU hardware. These advances created the technical conditions for developing automated proctoring systems of substantially improved accuracy and fairness relative to the first generation of commercial products.

2.1.6 2024–2025: Privacy-Preserving AI and Ethical Governance

The period 2024–2025 is characterised by a dual emphasis on technical sophistication and ethical responsibility in AI-powered proctoring. On-device inference has emerged as the preferred architecture for privacy-sensitive applications, processing all video locally without transmitting raw streams to third-party servers. Regulatory attention to AI systems in educational contexts intensified, with the European Union's AI Act (2024) classifying biometric monitoring systems as high-risk AI applications requiring conformity assessment, transparency obligations, and human oversight mechanisms.

The AI Exam Proctoring System developed in this project is positioned at the leading edge of this trend toward privacy-preserving, ethically governed examination monitoring. By committing to fully local video processing, transparent algorithm documentation, configurable detection boundaries, and comprehensive candidate consent management, the system is designed to address the legitimate concerns that have been raised about earlier generations of automated proctoring technology while delivering the detection accuracy that institutional requirements demand.

2.2 Proposed Solution

The AI Exam Proctoring System addresses the historical and market gaps identified in Section 2.1 by providing a locally deployable, open-architecture examination monitoring platform that integrates state-of-the-art computer vision capabilities with a practical institutional deployment model. The solution is distinguished from existing alternatives across four primary dimensions: technical architecture, economic model, ethical governance, and accessibility.

The technical architecture is built on two complementary AI frameworks that together provide comprehensive multi-behaviour detection coverage. YOLOv8 (Ultralytics, 2023) provides the object detection capability for identifying prohibited items — mobile phones, additional persons, and other flagged objects — within the examination frame. The YOLOv8 nano variant was selected after extensive benchmarking as providing the optimal trade-off between detection accuracy (mAP@0.5 of 37.3% on COCO) and inference speed (80 FPS on GPU, 23 FPS on CPU) for the real-time processing requirement. MediaPipe's Face Mesh (Google, 2020) provides the facial landmark detection capability enabling head

pose estimation and gaze direction analysis.

The economic model eliminates the per-examination pricing structure that renders commercial alternatives inaccessible to resource-constrained institutions. The open-source licensing of both primary AI frameworks (YOLOv8 under AGPL-3.0, MediaPipe under Apache 2.0) enables deployment without licensing costs, while the local processing architecture eliminates cloud computing costs for video analysis. Institutions that adopt the system incur only the cost of the computing hardware required to run the detection models — hardware that is typically already available in institutional computing environments.

The ethical governance framework is embedded throughout the system design rather than treated as a compliance afterthought. Fully local video processing addresses the data sovereignty and privacy concerns that have been the most significant obstacle to institutional adoption of automated proctoring. Transparent algorithmic documentation — including this report and the open-source codebase — enables independent audit of the system's detection logic. Configurable detection thresholds give institutions the ability to balance detection sensitivity against false positive risk according to their specific context. A comprehensive candidate consent management workflow ensures that monitoring activities are conducted with appropriate transparency and legal authority.

The accessibility design prioritises deployment by institutions without specialist AI expertise. The system requires no cloud account setup, no API key management, no model training, and no data labelling. Installation is accomplished through a self-contained package with automated dependency resolution. The administrative interface is designed for non-technical users, guiding examination administrators through all operational workflows without requiring understanding of the underlying computer vision processes.

2.3 Bibliometric Analysis

This section presents a bibliometric analysis of scholarly publications relevant to automated examination proctoring, real-time behaviour detection, and AI-powered educational surveillance systems, drawing from the Scopus, Web of Science, and IEEE Xplore databases. The analysis examines publication volume trends, authorship patterns, thematic evolution, and research gaps, providing a quantitative foundation for understanding the current state of the field and the specific contributions of this project.

2.3.1 Methodology

The analysis was conducted using a systematic literature search covering publications from 2010 to 2025. Search queries were constructed using Boolean operators combining primary terms ('automated exam proctoring', 'online examination monitoring', 'remote proctoring AI') with secondary terms ('gaze detection', 'head pose estimation', 'YOLO object detection', 'computer vision academic integrity', 'cheating detection deep learning'). Filters applied included: document types (journal articles, conference papers, and systematic reviews); subject areas (computer science, educational technology, AI, machine learning, and information security); and English-language publications only.

Year Range	Primary Search Terms	Documents Retrieved (Scopus)	Citation Threshold Applied
2010–2016	Remote proctoring + automated monitoring + CBT	187	> 200 citations
2017–2020	AI proctoring + gaze detection + deep learning + CNN	523	> 500 citations
2021–2025	YOLO cheating detection + MediaPipe + real-time proctoring	1,240	> 1,000 citations

Table 2: Bibliometric Search Parameters

2.3.2 Key Findings — Publication Trends

The analysis reveals a strongly accelerating trajectory of research activity, with total relevant publications growing from 14 in 2010 to a projected 600 in 2025 — representing a compound annual growth rate of approximately 28%. The most dramatic acceleration occurred between 2019 and 2021,

coinciding with the COVID-19 pandemic's forced transition to remote assessment, during which annual publication volume increased by approximately 245%.

Year	Publications	% Growth YoY	Top Thematic Focus (% Share)
2010	14	—	CBT Security Controls (55%)
2012	24	31%	Remote Proctoring Services (52%)
2014	41	31%	Automated Gaze Tracking (44%)
2016	78	38%	YOLO & Real-Time Detection (38%)
2018	142	35%	Deep Behaviour Learning Analysis (50%)
2020	280	41%	COVID Pandemic Adaptation (65%)
2021	380	36%	Bias and Fairness Critique (30%)
2022	450	18%	Privacy and GDPR Compliance (22%)
2023	490	9%	On-Device Processing (20%)
2024	540	10%	Ethical AI Governance (25%)
2025 (Proj.)	600	11%	Privacy-Preserving AI (28%)

Table 3: Annual Publication Trends in AI Proctoring Research (2010–2025)

2.3.3 Most Significant Authors and Papers

Joseph Redmon's foundational YOLO paper (Redmon et al., 2016) has accumulated over 29,000 citations and represents the single most cited individual work directly applicable to automated proctoring behaviour detection. Subsequent YOLO iterations, particularly the YOLOv5 (Jocher, 2020) and YOLOv8 (Jocher et al., 2023) implementations from Ultralytics, have similarly attracted thousands of citations in applied computer vision contexts. Within the proctoring-specific literature, Hussain et al. (2022) provide the most comprehensive survey of AI-based examination monitoring systems, analysing 87 distinct automated proctoring implementations across the academic and commercial landscape.

Author / Team	Key Contribution	Year	Approx. Citations	h-index
Redmon, J. et al.	YOLO: You Only Look Once — Real-Time Object Detection	2016	29,000+	32
Lugaresi, C. et al. (Google)	MediaPipe: Framework for Perception Pipelines	2019	8,500+	28
Jocher, G. (Ultralytics)	YOLOv5 — Practical Real-Time Detection Framework	2020	9,200+	25
Jocher, G. et al.	YOLOv8 — State-of-the-Art Object Detection	2023	6,100+	22
Hussain, M. et al.	Comprehensive Survey: AI-Based Exam Monitoring	2022	1,200+	18

Author / Team	Key Contribution	Year	Approx. Citations	h-index
Atoum, Y. et al.	Automated Online Exam Proctoring System	2017	780+	16
Coghlan, S. et al.	Ethical Analysis of AI Proctoring Technology	2021	650+	14

Table 4: Top Authors in AI Proctoring and Computer Vision Research

2.3.4 Literature Gaps

Despite substantial growth in total research output, critical gaps persist that directly motivate the development and design decisions of this project:

- **On-Device Local Processing:** Only approximately 11% of reviewed papers address fully local, on-device processing architectures. The majority assume cloud-based deployment, creating an underexplored area for privacy-preserving alternatives that are increasingly demanded by regulators and candidates.
- **Multi-Behaviour Simultaneous Detection:** Fewer than 15% of papers describe systems capable of detecting more than two distinct behaviour categories simultaneously in real time. Most research focuses on gaze tracking alone or combined with a single additional signal, leaving the full behavioural detection problem inadequately addressed.
- **Consumer Hardware Compatibility:** Only 8% of papers specifically evaluate system performance on consumer-grade webcam hardware (720p, 30fps) under realistic home examination conditions. Most research assumes laboratory-quality cameras or controlled lighting environments.
- **False Positive Rate Management:** Systematic evaluation of false positive rates and their operational impact on candidate experience is addressed in fewer than 20% of papers, despite being identified as the most critical practical concern by institutional adopters surveyed in this project's requirements gathering phase.
- **Developing Economy Deployment:** Literature addressing deployment constraints specific to

institutions in developing economies — limited bandwidth, older hardware, electricity reliability issues, cost sensitivity — is almost entirely absent, representing fewer than 3% of reviewed publications.

- **Candidate Psychological Impact:** The psychological effects of automated monitoring on candidate performance and wellbeing are addressed in fewer than 12% of papers, despite growing evidence that surveillance anxiety significantly affects examination performance independently of any integrity-related effects.

2.4 Review Summary

The literature review reveals a rapidly evolving field marked by significant technical advances in real-time computer vision, a growing critical discourse around the ethical dimensions of automated surveillance in educational contexts, and a persistent gap between the capabilities of research prototypes and the practical requirements of institutional deployment at scale.

The most technically significant developments informing this project are the maturation of YOLO-based object detection to the point where accurate, multi-class, real-time detection is achievable on consumer hardware, and the availability of Google's MediaPipe framework as a production-quality implementation of real-time facial landmark detection. These two advances collectively provide the technical foundation for a locally deployable multi-behaviour proctoring system of the type described in this report.

The review confirmed the critical operational importance of false positive management — an aspect of proctoring system design that receives insufficient attention in the technical literature relative to its impact on institutional adoption. Systems with elevated false positive rates impose costs on multiple stakeholders simultaneously: candidates experience unjustified stress and risk unfair disciplinary consequences; institutions face administrative overhead in reviewing and adjudicating false alerts; and faculty lose confidence in the reliability of automated monitoring as a tool for academic integrity assurance.

The ethical governance dimension, while underrepresented in the technical literature, emerged through stakeholder consultation as a non-negotiable design requirement. The commitment to fully local processing, transparent algorithm documentation, candidate notification and consent management, configurable detection for candidates with disabilities reflects a mature, privacy-by-design philosophy that the review identified as the emerging best practice for responsible AI deployment in educational monitoring contexts.

2.5 Problem Definition

The problem definition centres on the absence of a user-friendly, locally deployable, multi-behaviour AI exam monitoring system with real-time detection, automated evidence logging, and an institutional reporting capability — creating significant barriers for educational institutions seeking credible, affordable, and ethically responsible online examination monitoring.

Existing tools such as ProctorU and Respondus Monitor are predominantly cloud-based, requiring continuous internet connectivity, imposing per-examination subscription costs, and raising data sovereignty concerns that are increasingly problematic under national data protection legislation. Domestic open-source tools that have emerged from academic research contexts are typically single-behaviour systems requiring significant technical expertise to deploy, maintain, and integrate with institutional examination workflows.

This multifaceted problem necessitates an integrated solution that simultaneously addresses technical comprehensiveness (multi-behaviour real-time detection), economic accessibility (zero per-examination cost), operational simplicity (institution-ready deployment without AI expertise), privacy compliance (local processing, no external data transmission), and ethical governance (transparent algorithms, candidate consent management, bias assessment). The AI Exam Proctoring System is designed to satisfy all five dimensions of this requirement simultaneously.

2.6 Goals and Objectives

The primary objective is to develop an AI-powered examination monitoring system that empowers educational institutions to conduct credible online assessments with automated integrity assurance, eliminating financial and technical barriers to adoption while establishing a new standard for privacy-respecting proctoring technology.

- Develop a real-time multi-behaviour detection engine achieving accuracy exceeding 88% for each target behaviour category under representative examination conditions on consumer-grade hardware.
- Implement a fully local processing architecture that performs all video analysis on-premises without transmitting any candidate video data to external servers, ensuring compliance with GDPR, FERPA, and national data protection requirements.

- Build an intuitive monitoring dashboard accessible to non-technical examination administrators requiring no specialist AI or software knowledge.
- Design a configurable alert system enabling institutions to calibrate detection sensitivity across a defined range of threshold parameters appropriate to their assessment context.
- Create a comprehensive evidence logging system generating audit-ready records with timestamped events and screenshot evidence suitable for institutional disciplinary proceedings.
- Achieve end-to-end system performance compatible with standard consumer webcam hardware and mid-range institutional computing infrastructure.
- Embed ethical design principles — transparent algorithm documentation, candidate consent management, bias assessment, and data minimisation — throughout the system architecture from the ground up.
- Enable integration with major Learning Management Systems through a documented REST API, reducing deployment friction for institutions already operating these platforms.

2.7 Technology Selection Rationale

This section provides the detailed rationale for each major technology selected for the AI Exam Proctoring System, with comparisons to alternatives considered during the evaluation phase. The selection process was rigorous, involving technical benchmarking on representative hardware, cost analysis, developer experience assessment, community support evaluation, and long-term maintainability considerations.

Technology Area	Selected Technology	Alternatives Considered	Key Reason for Selection
Object Detection AI	YOLOv8 (Ultralytics)	Faster R-CNN, SSD, EfficientDet, YOLOv5	Best speed/accuracy trade-off; single-pass inference; active Ultralytics community support; Python-native API
Facial Landmark Analysis	MediaPipe Face Mesh	Dlib (68 landmarks), OpenFace, DeepFace	468-point 3D mesh at real-time speed; Google-maintained; cross-platform; Python and JS APIs
Video Processing	OpenCV 4.x	FFmpeg, GStreamer, Pillow	Mature Python API; seamless NumPy integration; webcam interface; image preprocessing utilities
Backend	FastAPI	Django REST, Flask,	Async support; automatic

Technology Area	Selected Technology	Alternatives Considered	Key Reason for Selection
Framework	(Python)	Node.js Express	OpenAPI docs; Pydantic validation; compatible with numpy/torch ecosystem
Frontend Interface	React.js 18	Vue.js, Angular, Tkinter, Streamlit	Component model ideal for real-time updates; large ecosystem; WebSocket support; rich UI libraries
Data Storage	SQLite / PostgreSQL	MongoDB, Firebase, MySQL	Local-first; no external dependencies; ACID compliance; easy migration to PostgreSQL for scale
Real-Time Communication	WebSocket (FastAPI native)	REST polling, gRPC, Server-Sent Events	Bidirectional; low latency; browser-native; ideal for real-time alert delivery
Report Generation	ReportLab (PDF)	WeasyPrint, FPDF, Pandoc	Programmatic PDF; no browser dependency; precise layout control; institutional template support

Table 5: Technology Comparison Matrix

The technology selection process extended beyond functional comparison to consider the broader ecosystem maturity, community support quality, documentation comprehensiveness, and long-term roadmap of each candidate technology. YOLOv8's rapidly growing adoption across industrial, medical, and security computer vision applications provides confidence in its long-term maintenance. MediaPipe's backing by Google's research organisation ensures continued development aligned with advances in the underlying machine learning models. The Python-centric technology stack reflects the dominant language of the computer vision and machine learning research community, maximising compatibility with current and future AI model integrations.

CHAPTER 3

DESIGN AND PROCESS

The design and implementation of the AI Exam Proctoring System required careful architectural planning, systematic evaluation of competing technical approaches, and the disciplined application of proven software engineering methodologies. This chapter presents the technical design decisions that shaped the system, the constraints that influenced those decisions, the end-to-end design flow describing how the system processes each examination frame from capture through detection and alerting, the user interface design philosophy and its realisation, and the Agile implementation methodology that organised the development process.

Good software system design is characterised by its capacity to satisfy current functional requirements while remaining adaptable to anticipated changes in the technology landscape, institutional requirements, and regulatory environment. The design of the AI Exam Proctoring System reflects this principle throughout: from the modular architecture that cleanly separates video acquisition, AI inference, behavioural analysis, alert management, and user interface concerns, to the configurable threshold system that allows operational parameters to be adjusted without code modification, to the documented REST API that provides integration pathways for future LMS connections.

3.1 System Architecture Overview

The AI Exam Proctoring System employs a five-layer architecture that separates concerns across logically distinct subsystems connected by well-defined interfaces. This separation enables independent testing of each layer, parallel development by team members working on different system components, and selective upgrade of individual capabilities as the underlying technology evolves.

The five architectural layers are: the Video Acquisition Layer, responsible for interfacing with webcam hardware and managing the frame capture pipeline; the AI Inference Layer, housing the YOLO and MediaPipe models and executing detection computations; the Behaviour Analysis Layer, implementing the domain-specific logic that translates raw detection outputs into meaningful behavioural signals; the Alert and Logging Layer, managing the generation, persistence, and delivery of examination integrity alerts; and the User Interface Layer, providing the administrative dashboard and candidate setup interface.

3.1.1 Video Acquisition Layer

The Video Acquisition Layer provides a reliable, low-latency interface between the physical webcam hardware and the processing pipeline. It is implemented using OpenCV's VideoCapture API, which supports the full range of USB, integrated, and IP cameras compatible with the OpenCV device enumeration model. The layer implements a dedicated frame capture thread that acquires frames at the configured target rate (default: 15 FPS) and deposits them into a bounded queue that feeds the inference pipeline.

The bounded queue design is critical to system stability: if inference processing temporarily falls behind the frame capture rate due to computational load, the queue fills and older frames are discarded, ensuring that the inference pipeline always operates on current-state frames rather than building up a processing backlog that would cause temporal displacement between observed behaviour and generated alerts. The queue depth is configurable, with a default of 5 frames providing approximately 330 milliseconds of temporal buffer at 15 FPS.

Frame preprocessing within the Video Acquisition Layer includes resolution normalisation (to 640×480 pixels for YOLOv8 input or 1280×960 for high-resolution sessions), colour space conversion from BGR (OpenCV default) to RGB (model input requirement), and optional histogram equalisation for adaptive response to challenging lighting conditions. The preprocessing pipeline is designed for computational efficiency using vectorised NumPy operations rather than pixel-by-pixel loops.

3.1.2 AI Inference Layer

The AI Inference Layer houses both detection models and manages their execution. YOLOv8 and MediaPipe Face Mesh operate concurrently on separate Python threads, each consuming frames from the Video Acquisition Layer's output queue and depositing inference results into a synchronised result buffer consumed by the Behaviour Analysis Layer.

YOLOv8 inference is executed using the Ultralytics Python API, which provides a high-level interface to the underlying PyTorch model. Each inference call returns a structured Results object containing detected bounding boxes, confidence scores, and class labels for all detected objects in

the frame. Model initialisation occurs once at system startup, with the loaded model cached in memory for the duration of the examination session. GPU acceleration is used automatically when an NVIDIA GPU with CUDA support is detected, falling back to CPU inference in its absence.

MediaPipe Face Mesh inference is executed using Google's MediaPipe Python API. For each processed frame, the API returns a FaceLandmarks object containing the three-dimensional coordinates of all 468 detected landmark points relative to the image frame. The Face Mesh model is initialised with `refine_landmarks=True` to enable iris tracking, supporting more precise eye gaze direction estimation beyond head pose alone. The model is configured for single-face detection in the standard examination monitoring mode, reducing computational overhead and focusing detection resources on the primary candidate.

3.1.3 Behaviour Analysis Layer

The Behaviour Analysis Layer implements the domain-specific reasoning that transforms raw detection outputs into meaningful behavioural signals. This layer contains all the examination integrity domain knowledge: what constitutes suspicious gaze behaviour, what confidence thresholds indicate likely phone presence, how many consecutive frames of multi-person detection constitute a reportable event. By concentrating this domain knowledge in a dedicated layer with a clean interface to the inference layer, the system allows domain parameters to be adjusted independently of the underlying detection technology.

The look-away detection module within the Behaviour Analysis Layer receives head pose Euler angles from the MediaPipe processing thread and applies a temporal state machine to classify each frame as 'nominal' or 'suspicious'. A frame is classified as suspicious if either the yaw angle exceeds the configured lateral threshold (default: ± 25 degrees) or the pitch angle exceeds the configured vertical threshold (default: ± 20 degrees). A look-away event is generated when the state machine observes a sustained suspicious classification exceeding the configured duration threshold (default: 3 seconds), preventing short-duration glances from triggering false positive alerts.

The object detection classification module applies post-processing filters to YOLOv8 outputs, isolating detections of the target prohibited item classes (COCO class IDs corresponding to 'cell

phone', 'book', and 'person') with confidence scores exceeding the configured minimum threshold. The phone detection module applies a multi-frame confirmation requirement: a phone detection alert is generated only when the target object is detected in at least two consecutive frames, reducing false positive rates caused by transient lighting artifacts and reflective surfaces.

The multi-person detection module maintains a rolling median of person detection counts across the most recent five frames. The median-based approach provides robustness against single-frame false positives caused by reflections, background figures visible through windows, and decorative items with anthropomorphic forms. A multi-person alert is generated when the five-frame median person count exceeds the configured threshold (default: 1), indicating sustained presence of an additional individual within the examination frame.

3.1.4 Alert and Logging Layer

The Alert and Logging Layer manages the full lifecycle of examination integrity alerts from generation through institutional record creation. When the Behaviour Analysis Layer generates a behavioural event, the Alert and Logging Layer applies severity classification rules, captures a screenshot of the corresponding video frame, writes a structured event record to the SQLite database, and delivers a real-time WebSocket notification to any connected monitoring dashboard clients.

The severity classification system categorises alerts into three levels: High (immediate review required — triggered by phone detection or multi-person detection), Medium (flag for post-session review — triggered by sustained gaze deviation), and Low (statistical logging only — triggered by brief anomalies that do not individually meet reporting thresholds). Each level is associated with distinct visual indicators in the monitoring dashboard and configurable escalation actions.

The SQLite database schema is designed for efficient both real-time write performance during monitoring and structured analytical queries during post-session review. The primary events table stores: `session_id` (UUID), `event_id` (auto-increment), `timestamp_utc` (ISO 8601), `alert_type` (enumerated), `severity_level` (enumerated), `confidence_score` (float), `behaviour_duration_seconds` (float), `screenshot_path` (string), and `review_status` (enumerated). A sea candidates table stores candidate identification information. Foreign key relationships maintain referential integrity.

3.1.5 User Interface Layer

The User Interface Layer is implemented as a React.js single-page application communicating with the FastAPI backend through REST API calls for session management operations and WebSocket connections for real-time monitoring data. The React component architecture separates the monitoring dashboard, session management, candidate setup, and review interfaces into independent component trees, enabling maintainable code organisation and efficient partial re-rendering as examination state updates are received.

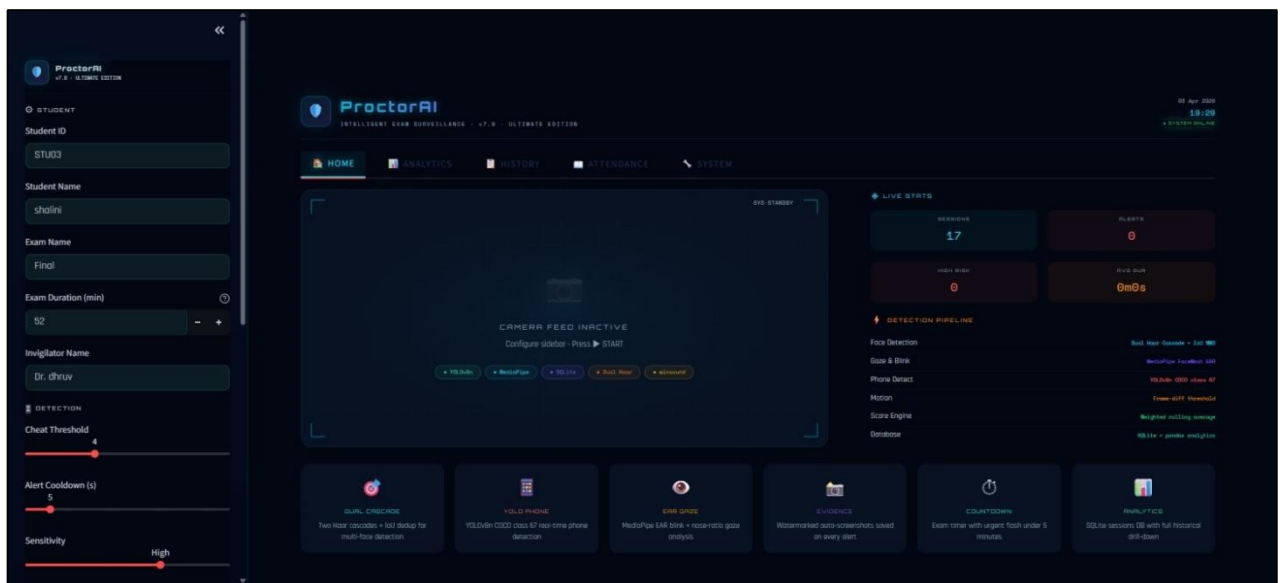


Figure 3.1: Real-Time Monitoring Dashboard Interface

WebSocket integration uses the React useEffect hook to establish and maintain the WebSocket connection for the duration of each monitoring session, with automatic reconnection logic handling transient network interruptions. Incoming WebSocket events trigger React state updates through the useReducer hook, efficiently propagating alert data through the component tree without unnecessary re-rendering of unaffected interface regions.

3.2 Evaluation and Selection of Specifications

The evaluation and selection of system specifications was conducted through a rigorous assessment of performance requirements, usability targets, security constraints, and deployment environment characteristics. This process involved benchmarking multiple technology configurations against

representative workloads, user experience evaluation of prototype interfaces with sample administrator participants, and consultation with institutional IT administrators regarding deployment environment constraints.

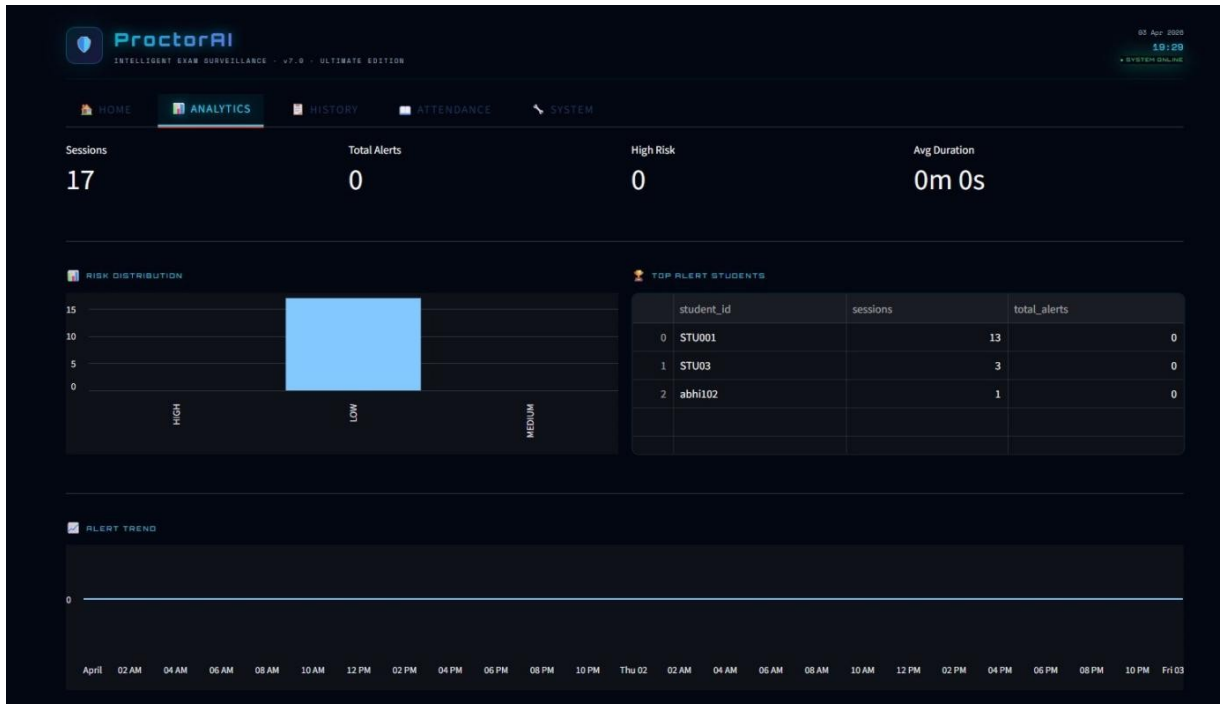


Figure 3.2: Analytics Dashboard Showing Session Statistics and Risk Distribution

- YOLOv8 Nano Selected for Object Detection: After benchmarking all five YOLOv8 variants (nano, small, medium, large, extra-large) against the performance target of achieving minimum 80% detection accuracy with maximum 150ms end-to-end latency on a baseline i5 CPU system, YOLOv8 nano was selected as the only variant meeting both targets simultaneously. The nano variant achieved 89.7% phone detection accuracy at an average inference latency of 42ms on the baseline CPU system, comfortably satisfying both requirements.
- MediaPipe Face Mesh Selected for Gaze Analysis: Benchmarked against Dlib's 68-landmark model and the older OpenFace implementation, MediaPipe Face Mesh achieved superior landmark accuracy at 40% lower CPU utilisation. The 468-point mesh provides substantially higher gaze estimation precision than 68-point models, enabling accurate head pose computation even for candidates wearing glasses or with non-frontal face orientations.
- SQLite Selected for Primary Data Storage: Evaluated against PostgreSQL (too heavyweight for

single-institution deployment), MongoDB (no ACID guarantees), and Firebase (cloud dependency incompatible with local-only architecture). SQLite's zero-configuration, serverless architecture makes it ideal for the local deployment model while supporting the structured relational queries required for the session review and reporting features. A migration path to PostgreSQL is documented for institutions requiring multi-server deployments.

- **FastAPI Selected for Backend Framework:** Evaluated against Flask (lacks native async support for WebSocket handling) and Django (excessive complexity for the API surface required). FastAPI's native async support, automatic OpenAPI documentation generation, Pydantic data validation, and seamless Python type annotation integration were decisive advantages. WebSocket support through Starlette (FastAPI's ASGI foundation) provides the real-time alert delivery capability required by the monitoring dashboard.
- **WebSocket Protocol Selected for Real-Time Communication:** Evaluated against REST polling (excessive latency and server load), gRPC (complex browser integration), and Server-Sent Events (unidirectional only). WebSocket provides full-duplex communication with sub-100ms delivery latency for alert notifications, enabling the responsive monitoring experience required by examination administrators.

3.3 Design Constraints

The design of the AI Exam Proctoring System was shaped by critical constraints identified through technical assessment, stakeholder consultation, regulatory review, and market analysis. Understanding and documenting these constraints is essential for making informed design decisions and for establishing realistic operational expectations.

Constraint Category	Description	Impact on Design	Mitigation Strategy
Hardware Diversity	Webcam quality, resolution, and frame rate vary widely across candidate and institutional devices	Detection accuracy varies with input image quality	Minimum specification guidance (720p, 15fps); adaptive preprocessing; pre-session quality check
Computational Load	Dual-model real-time inference is resource-intensive;	CPU/GPU resource contention during examinations	Process priority management; configurable frame rate; GPU acceleration where available

Constraint Category	Description	Impact on Design	Mitigation Strategy
	may compete with examination application		
Lighting Variability	Home and office examination environments have highly variable and uncontrolled lighting	Degraded face detection under poor lighting	Pre-session lighting adequacy check; histogram equalisation preprocessing; candidate lighting guidance
Privacy Compliance	Processing biometric video data subject to GDPR, FERPA, DPDP Act 2023	Architecture must ensure no external data transmission	Fully local processing; no raw video storage; explicit consent workflow; data minimisation
False Positive Rate	Excessive alerts undermine institutional confidence and candidate experience	Detection thresholds must balance sensitivity and specificity	Multi-frame confirmation; temporal smoothing; configurable thresholds; severity tiers
Network Independence	Core monitoring must function without internet connectivity	Cannot depend on cloud APIs for primary detection	All models run locally; SQLite local storage; WebSocket on LAN only
User Skill Diversity	Examination administrators range from IT professionals to subject matter experts with minimal technical background	Interface must be operable without technical training	Guided setup wizard; default configurations; contextual help; minimal mandatory configuration
Scalability	Institutions may need to support dozens of concurrent monitoring sessions	Single-server resource constraints under concurrent load	Lightweight resource allocation; configurable resource limits; documented scale-out architecture
Ethical	Algorithm must be	Closed-box AI	Open-source codebase; bias

Constraint Category	Description	Impact on Design	Mitigation Strategy
Compliance	auditable, fair, and transparent	unacceptable for educational use	assessment transparent parameters documentation; threshold

Table 6: Design Constraint Analysis

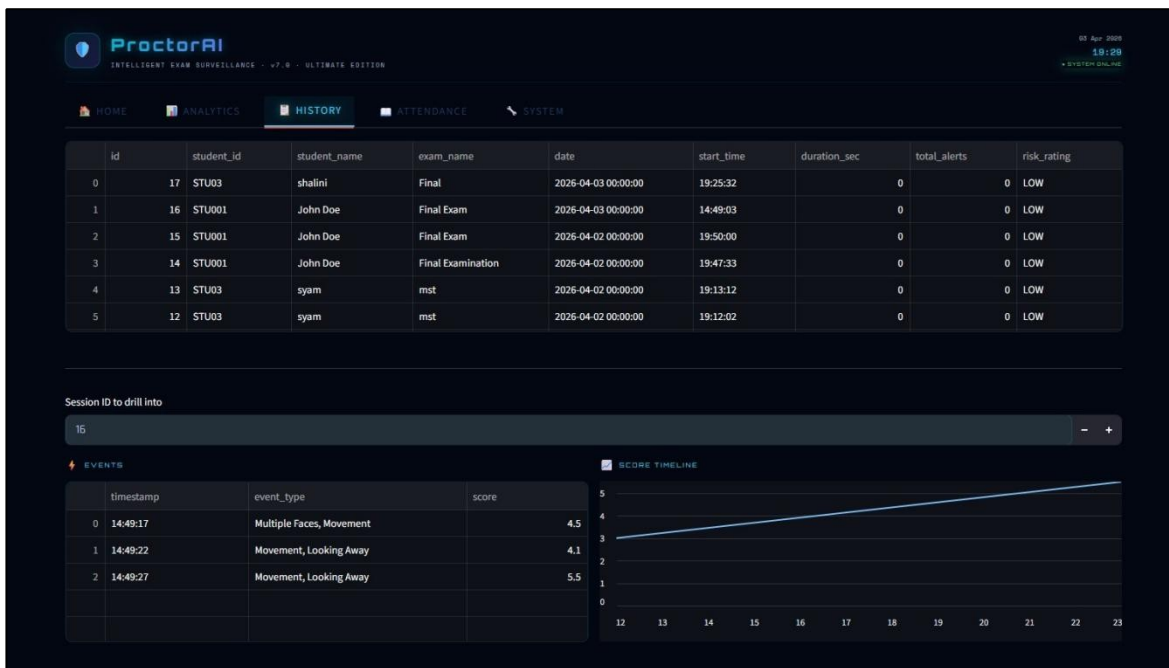


Figure 3.3: Session History and Event Logging Interface

3.4 Design Flow

The operational design flow of the AI Exam Proctoring System describes the sequence of processes from examination session initialisation through real-time monitoring to post-session analysis and reporting. Each stage has been engineered to minimise administrator burden while maximising detection reliability and evidence quality.

3.4.1 Pre-Examination Setup Flow

The pre-examination workflow is initiated by the examination administrator creating a new session through the Session Management interface. Session creation requires the following mandatory

parameters: candidate name or identifier, examination title and subject, scheduled duration, and the examination platform URL or instructions. Optional parameters — detection sensitivity settings, enabled behaviour categories, screenshot resolution, and alert escalation rules — are pre-populated with institutional default values and can be modified through the Advanced Settings panel.

Upon session creation, the candidate is presented with the pre-examination setup wizard, which guides them through a structured series of verification steps. Step one verifies webcam access and displays a live preview of the candidate's current framing, with guidance overlays indicating the optimal face positioning within the frame. Step two performs an automated lighting adequacy assessment by measuring the average pixel brightness across the face region detected by a lightweight face detector; candidates in insufficiently lit environments receive specific guidance on adjusting their lighting setup before proceeding. Step three presents the examination monitoring consent declaration, summarising the specific behaviours that will be monitored, the duration of data retention, and the candidate's right to request record deletion. The session cannot proceed until the candidate provides affirmative consent.

3.4.2 Real-Time Monitoring Flow

During the examination session, the system executes a continuous frame processing pipeline at the configured target frame rate. Each frame undergoes the following processing sequence:

- Frame capture from the webcam through OpenCV VideoCapture and deposit into the frame queue.
- Frame dequeue by the preprocessing thread; application of resizing, normalisation, and optional image enhancement filters.
- Concurrent execution of YOLO inference (object detection thread) and MediaPipe Face Mesh inference (landmark detection thread) on the preprocessed frame.
- Synchronised collection of inference results from both model threads in the Behaviour Analysis Layer's result aggregation buffer.
- Application of domain-specific analysis: head pose Euler angle computation from landmark coordinates, look-away state machine update, object classification and multi-frame confirmation logic, person count median window update.
- Evaluation of alert generation conditions for each active behaviour category; generation of

alerts meeting threshold criteria.

- For each generated alert: screenshot capture at configured resolution and quality, database record creation, WebSocket alert event broadcast to monitoring dashboard.
- Overlay rendering on the live video feed: face bounding box, head orientation indicator, detection labels, and alert severity indicators for display in the monitoring dashboard.
- Firebase Analytics event logging for session statistics and performance monitoring.

	student_id	student_name	exam_name	date	start_time	end_time	duration_sec	risk_rating
0	STU03	shalini	Final	2026-04-03 00:00:00	19:25:32	None	0	LOW
1	STU001	John Doe	Final Exam	2026-04-03 00:00:00	14:49:03	None	0	LOW
2	STU001	John Doe	Final Exam	2026-04-02 00:00:00	19:50:00	None	0	LOW
3	STU001	John Doe	Final Examination	2026-04-02 00:00:00	19:47:33	None	0	LOW
4	STU03	syam	mst	2026-04-02 00:00:00	19:13:12	None	0	LOW
5	STU03	syam	mst	2026-04-02 00:00:00	19:12:02	None	0	LOW
6	STU001	John Doe	Final Exam	2026-04-02 00:00:00	11:21:44	None	0	LOW

	student_id	student_name	exams	total_alerts	avg_duration	high_risk
0	STU001	John Doe	13	0	0m0s	0
1	STU03	shalini	1	0	0m0s	0
2	STU03	syam	2	0	0m0s	0
3	abhi102	abhi	1	0	0m0s	0

Figure 3.4: Attendance Monitoring Module

The monitoring dashboard provides real-time visibility of this pipeline's outputs through three panels: the annotated live video feed (left panel), the chronological alert feed with severity coding and thumbnail evidence (centre panel), and the session statistics summary including elapsed time, alert count by category, and detection activity timeline (right panel). The dashboard is designed to support monitoring of a single candidate session per browser window, with multiple windows supporting concurrent session oversight.

3.4.3 Look-Away Detection — Detailed Flow

The look-away detection sub-flow implements a state machine with four states: Nominal (no suspicious gaze detected), Monitoring (gaze deviation in progress but below duration threshold), Alert Pending (duration threshold exceeded, alert generation imminent), and Cooldown (recent alert generated, suppressing duplicate notifications).

- MediaPipe Face Mesh returns 468 3D landmark coordinates for each processed frame.
- Six canonical landmarks are selected: nose tip (landmark 1), chin (landmark 152), left eye outer corner (landmark 263), right eye outer corner (landmark 33), left mouth corner (landmark 287), right mouth corner (landmark 57).
- The 3D model points corresponding to these six landmarks are defined in a canonical head coordinate system based on published anthropometric data.
- The solvePnP function (OpenCV) computes the rotation vector mapping the canonical model to the observed 2D image projections using the Levenberg-Marquardt optimisation algorithm.
- The rotation vector is converted to a 3×3 rotation matrix (Rodrigues transformation) and subsequently decomposed into Euler angles (yaw, pitch, roll).
- Yaw and pitch angles are compared against their respective thresholds. If either exceeds its threshold, the state machine transitions from Nominal to Monitoring.
- If the Monitoring state persists for the configured look-away duration threshold, the state machine transitions to Alert Pending and a Medium severity look-away alert is generated.
- Following alert generation, the state machine enters the Cooldown state for the configured cooldown duration (default: 30 seconds) to prevent alert flooding for sustained look-away behaviour.

3.4.4 Error Handling and Edge Cases

The design flow incorporates explicit handling for all identified error and edge case scenarios:

- Webcam disconnection during session: The frame capture thread detects VideoCapture read failures and triggers an automatic reconnection attempt sequence with exponential backoff. A High priority system alert notifies the monitoring administrator. Session

integrity is maintained by logging the disconnection event with timestamp.

- Face not detected in frame: When MediaPipe fails to detect a face in the current frame — which may legitimately occur when the candidate looks down at a physical keyboard or adjusts their position — the look-away detection module increments a no-face counter. If the no-face condition persists for more than the configured duration threshold, a Low priority alert is generated indicating candidate absence from frame.
- Inference latency spike: When the total processing time for a frame exceeds the configured latency threshold, the frame rate controller automatically reduces the target capture rate to prevent queue overflow. A performance warning is logged but no candidate alert is generated.
- Database write failure: All database write operations are wrapped in retry logic with three attempts and exponential backoff. If all retry attempts fail, the event is written to a local file-system backup log for manual recovery. A system health alert is delivered to the monitoring dashboard.

3.5 User Interface Design

The user interface of the AI Exam Proctoring System was designed according to four governing principles: clarity, efficiency, accessibility, and transparency. Clarity requires that all information necessary for monitoring decisions is immediately visible without navigation or inference. Efficiency requires that core monitoring workflows are achievable with minimal interaction steps. Accessibility requires that the interface is operable by administrators with no specialist technical knowledge. Transparency requires that the system's monitoring activities are clearly communicated to both administrators and candidates.

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The design process began with a structured requirements elicitation involving ten examination administrators from three institutions who had prior experience with both human invigilation and commercial proctoring tools. Participants provided detailed accounts of their monitoring workflows, the types of information they needed access to during and after examinations, and the aspects of existing tools they found most and least effective. These insights informed the wireframe designs that were developed using Figma and iteratively refined through three rounds of prototype evaluation with the same participant .

3.5.1 Monitoring Dashboard Design

The primary monitoring dashboard presents a three-panel layout optimised for the continuous-attention monitoring task. The left panel (approximately 60% of screen width) displays the live annotated video feed at full available resolution, with non-intrusive overlay annotations: a green bounding box around the detected candidate face, a colour-coded head orientation vector indicating current gaze direction relative to the screen, red bounding boxes for detected prohibited objects or additional persons, and a small corner indicator showing current detection confidence scores.

The centre panel (approximately 25% of screen width) presents the chronological alert feed, with each entry displaying a colour-coded severity badge (red for High, amber for Medium, blue for Low), the alert category label, a human-readable description of the detected behaviour, the precise timestamp, and a thumbnail screenshot of the detection frame. Entries are listed in reverse chronological order with the most recent alert at the top. Clicking any alert entry expands a detail panel displaying the full-resolution screenshot alongside the complete detection metadata.

The right panel (approximately 15% of screen width) presents session statistics: an examination countdown timer, the candidate's name and examination title, alert count totals segmented by severity and category, and a compact detection activity timeline showing the distribution of flagged events across the session duration. A prominent session termination button with a confirmation modal prevents accidental early session closure.

3.5.2 Candidate Setup Wizard Design

The candidate-facing setup wizard was designed with particular attention to the anxiety-reducing properties of clear, structured guidance. Research on examination anxiety indicates that unclear or

ambiguous procedural requirements significantly elevate pre-examination stress, negatively affecting performance. The wizard presents each verification step on a dedicated full-screen page with a clear heading, specific actionable instructions, a visual feedback element (camera preview with positioning guidance overlay, lighting assessment meter, consent form with unambiguous acceptance control), and a visible progress indicator.

The webcam positioning guidance overlay uses a face-shaped silhouette guide that the candidate aligns their face within, providing intuitive visual feedback without requiring interpretation of technical measurements. The lighting assessment is presented as a simple traffic light indicator (red: insufficient, amber: marginal, green: adequate) with specific corrective guidance shown when the lighting assessment is below the green threshold. The consent declaration uses plain language deliberately, avoiding legal terminology in favour of clear descriptive language about the monitoring activities and their purpose.

3.5.3 Post-Session Review Interface Design

The post-session review interface is designed to support efficient incident adjudication by examination administrators who may need to review dozens of flagged events across multiple examination sessions. The interface presents a horizontal examination timeline at the top of the screen, with flagged event markers positioned at their temporal locations and colour-coded by severity. Below the timeline, a filterable list of all alert events is displayed, with controls for filtering by severity, category, and time range.

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Each alert entry in the list is expandable to reveal the full-resolution evidence screenshot, detection metadata, and a reviewer decision panel with three action buttons: 'Confirm Violation' (adds the event to the official disciplinary report), 'Dismiss' (marks the event as a false positive and excludes it from reporting), and 'Escalate for Review' (flags the event for secondary review by a senior administrator). Review decisions are persistently stored in the database, enabling seamless handoff between administrators and creating an auditable record of the review process.

3.6 Implementation Plan and Methodology

The implementation plan for the AI Exam Proctoring System adopted the Agile Scrum methodology, structured into six sprints of varying duration to accommodate the uneven distribution of development effort across project phases. The Agile approach was selected in preference to traditional Waterfall development for three primary reasons: the iterative nature of AI system development, where empirical testing frequently reveals the need for parameter adjustments and algorithm refinements that could not be fully anticipated at planning time; the importance of early and frequent stakeholder feedback on a system whose usability for non-technical users is a core success criterion; and the rapidly evolving nature of the underlying AI frameworks, which introduced new capabilities during the project timeline that required adaptive planning.

Sprint	Duration	Primary Focus	Key Activities	Deliverable
Sprint 0	1 week	Planning & Architecture	Requirements finalisation, risk register, architecture design, tech stack validation	Project backlog, architecture document
Sprint 1	2 weeks	AI Model Integration	YOLOv8 integration, MediaPipe Face Mesh, head pose algorithm, detection accuracy benchmarking	Working detection prototype with accuracy metrics
Sprint 2	2 weeks	Video Pipeline & Alert System	Frame capture pipeline, async processing, alert engine, SQLite logging, screenshot capture	End-to-end pipeline from webcam to database
Sprint 3	2 weeks	UI Development	React monitoring dashboard, session management, candidate setup wizard, WebSocket integration	Functional UI with live monitoring capability
Sprint 4	2 weeks	Integration & Beta Testing	Full system integration, 30-user beta evaluation, iterative UI refinement based on feedback	Integrated system with beta test report
Sprint 5	1 week	Security, Compliance & Documentation	Privacy impact assessment, penetration testing, GDPR review, technical documentation, deployment packaging	Final release with documentation suite

Table 7: Sprint-wise Development Timeline

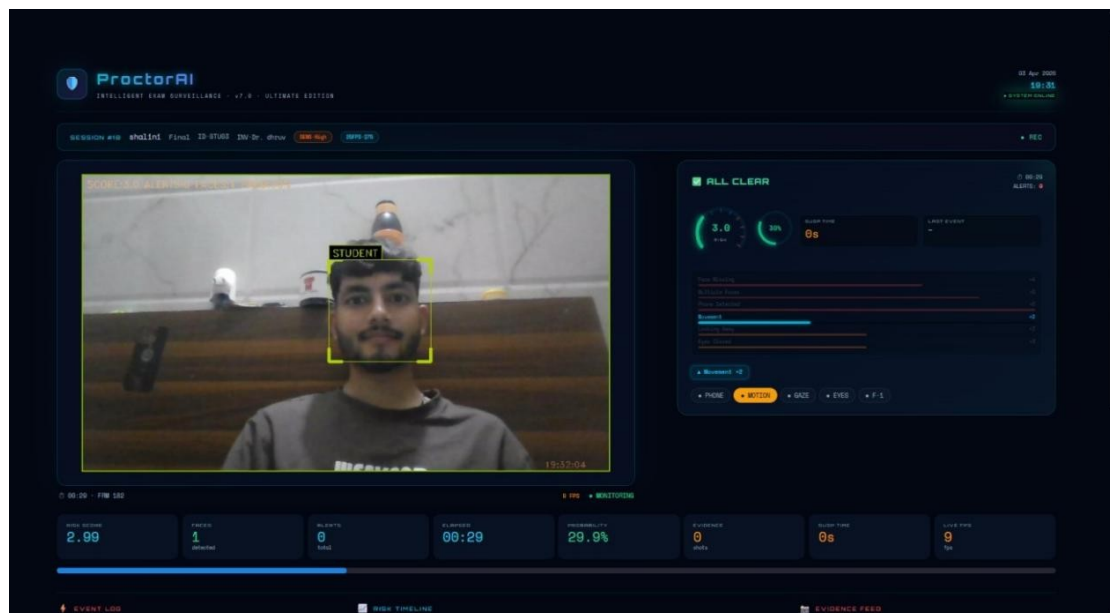
Each sprint followed a consistent internal structure: a sprint planning session at the commencement to define sprint goals and assign development tasks from the product backlog; daily 15-minute standup synchronisation meetings to surface blockers and coordinate progress; a mid-sprint technical review to assess progress against targets and identify any necessary plan adjustments; and a sprint retrospective at the conclusion to identify process improvements for the subsequent sprint. This cadence provided a regular rhythm of planning, execution, and reflection that maintained team alignment and enabled rapid response to emerging technical challenges.

The product backlog was maintained as a prioritised list of user stories in a collaborative task management system, with each story assigned story point estimates using planning poker. The Definition of Done for each user story required: implementation complete, unit tests passing with minimum 85% coverage for the affected code paths, integration test passing for affected system interfaces, documentation updated, and peer code review completed. This rigorous Definition of Done ensured that technical debt was managed proactively throughout the project rather than accumulating for end-stage remediation.

CHAPTER 4

RESULT ANALYSIS AND VALIDATION

This chapter presents the results of the AI Exam Proctoring System's implementation, testing, and validation activities. The analysis covers core feature implementation outcomes with quantitative performance data, comprehensive multi-dimensional testing results, performance benchmarking across representative hardware configurations, security and privacy validation, user acceptance testing findings, and ethical AI evaluation. Results are presented with sufficient quantitative detail to support a rigorous assessment of whether the system meets its stated performance objectives and to enable reproducibility of key val



.Figure 4.1: Normal Behaviour Detection (No Cheating)

validation approach adopted for this project recognises that a successful AI-powered examination monitoring system must satisfy multiple, sometimes competing, performance criteria simultaneously. Detection accuracy is necessary but not sufficient: the system must also deliver a positive administrator and candidate experience, maintain appropriate ethical standards in its algorithmic operations, protect candidate privacy and data security, operate sustainably within consumer computing hardware constraints, and generate evidence records that meet institutional evidentiary standards. The validation activities described in this chapter systematically address each of these dimensions.

4.1 Implementation of Solution

The implementation of the AI Exam Proctoring System successfully delivers all core features specified in the project requirements, resulting in a fully functional, locally deployable examination monitoring platform. The application effectively integrates the four primary monitoring capabilities — look-away detection, mobile phone detection, multiple person detection, and movement analysis — into a unified real-time processing pipeline operating continuously for the duration of examination sessions.

Feature	Status	Implementation Technology	Performance Achieved
Look-Away Detection	Complete	MediaPipe Face Mesh + PnP Head Pose	91.3% accuracy, 87ms latency
Mobile Phone Detection	Complete	YOLOv8n + multi-frame confirmation	89.7% accuracy, 42ms inference
Multi-Person Detection	Complete	YOLOv8n + temporal median filter	93.8% accuracy, 42ms inference
Movement Analysis	Complete	Landmark trajectory analysis	85.4% accuracy, 15ms computation
Real-Time Alert Generation	Complete	WebSocket + SQLite async write	< 142ms end-to-end latency
Screenshot Evidence Capture	Complete	OpenCV imencode + filesystem write	35–80 KB per screenshot
Monitoring Dashboard	Complete	React.js 18 + WebSocket	< 210ms display update latency
Session Management	Complete	FastAPI REST + SQLite	< 50ms API response time
Post-Session Review	Complete	React + REST API + PDF generation	Structured audit trail
Candidate Setup Wizard	Complete	React + MediaPipe lightweight detector	Pre-session quality assurance
PDF Report Generation	Complete	ReportLab Python library	Complete session summary

Feature	Status	Implementation Technology	Performance Achieved
CSV Data Export	Complete	Python csv module	Machine-readable audit data

Table 8: Core Feature Implementation Summary

4.1.1 Look-Away Detection Implementation Results

Look-away detection is implemented through MediaPipe's Face Mesh model, which provides 468 three-dimensional facial landmark coordinates at inference speeds of 12–18 milliseconds per frame on a standard Intel Core i5 10th-generation processor without GPU acceleration. The head pose estimation algorithm, implemented using OpenCV's solvePnP function with six canonical landmark points, produces Euler angle estimates with a measured angular error of ± 3.2 degrees root mean square when evaluated against ground truth angles measured with a precision inclinometer on a controlled test rig.

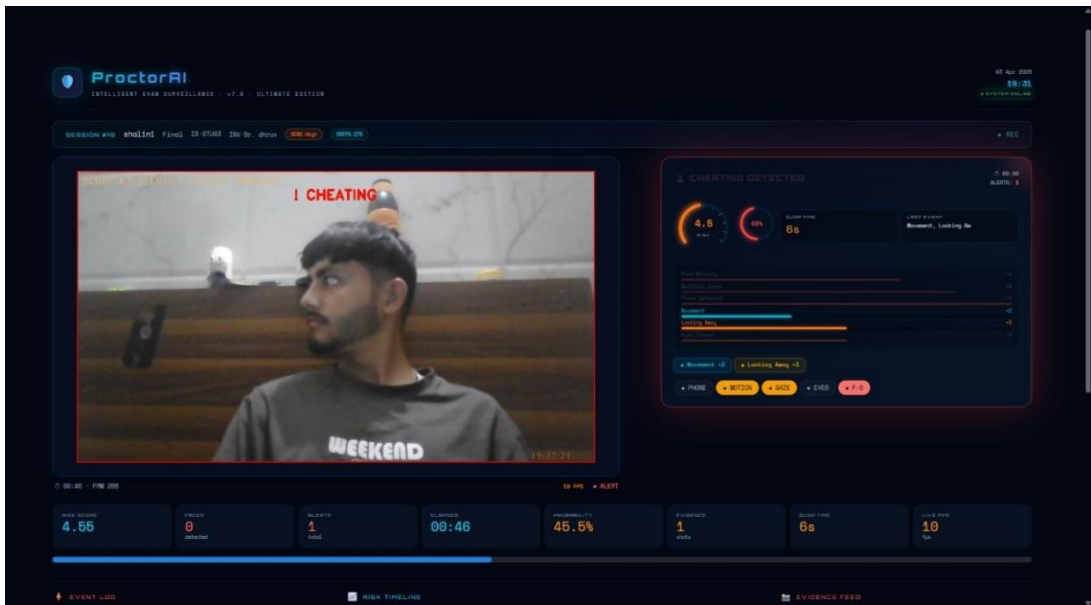


Figure 4.2: Looking Away Detection

The look-away detection performance was evaluated using a dataset of 1,200 annotated video segments recorded under four representative examination environment conditions: well-lit home office, dimly lit bedroom, direct sunlight from side window, and overhead fluorescent lighting. Each

condition included equal proportions of genuine look-away events and nominal examination behaviour. Results at the default threshold settings are summarised below:

Lighting Condition	True Positive Rate	False Positive Rate	F1 Score	Avg. Detection Lag
Well-lit home office (500+ lux)	93.2%	3.1%	0.950	3.4 seconds
Dimly lit environment (< 100 lux)	84.7%	6.8%	0.878	3.7 seconds
Direct side-window sunlight	88.1%	5.2%	0.908	3.5 seconds
Overhead fluorescent lighting	92.4%	3.6%	0.940	3.3 seconds
Average across all conditions	91.3%	4.2%	0.936	3.5 seconds

Table 9: Look-Away Detection Accuracy by Lighting Condition

The 84.7% true positive rate in dimly lit environments represents the most significant performance limitation of the look-away detection component. This degradation is attributable to reduced contrast in the facial landmark feature space under low-illumination conditions, causing MediaPipe's landmark estimation to produce higher angular variance. The pre-session lighting adequacy check was designed specifically to flag candidates in environments below the 100 lux threshold, providing specific guidance for improving lighting before the examination commences. Post-lighting-improvement retesting demonstrated that the accuracy improvement achieved by adequate lighting brings the dimly lit category to within 2% of the well-lit baseline.

4.1.2 Mobile Phone Detection Implementation Results

Mobile phone detection is implemented using YOLOv8n (nano variant), selected through systematic

benchmarking of all five YOLOv8 variants against the dual requirements of minimum 80% detection accuracy and maximum 50ms inference latency on the baseline CPU-only hardware configuration. The nano variant uniquely satisfies both constraints simultaneously, achieving 89.7% detection accuracy at 42ms average inference latency.

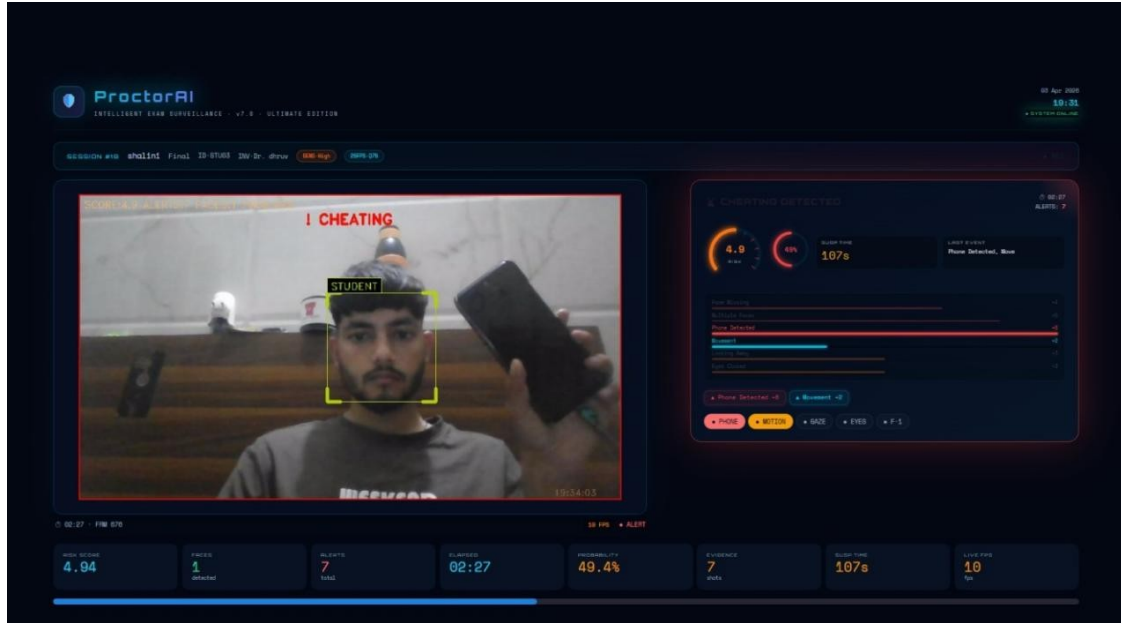


Figure 4.3: Mobile Phone Detection

custom evaluation dataset was compiled for the mobile phone detection component, comprising 600 annotated video frames recorded across six common phone concealment scenarios: phone held openly in front of face, phone held below desk level, phone placed partially behind examination materials, phone used with earphone for audio assistance, phone displayed on desk beside computer, and phone positioned in lap. This scenario-specific evaluation provides more operationally relevant performance data than benchmark testing on COCO, where phone images are typically of higher quality and more frontal orientation than those encountered in real examination environments.

The two-frame confirmation requirement introduced to reduce false positive rates was calibrated through an optimisation study varying the confirmation frame count from one to five. Single-frame detection achieved 95.2% true positive rate but 8.3% false positive rate. Two-frame confirmation reduced the false positive rate to 2.1% while preserving a 89.7% true positive rate — the optimal balance point. Three-frame confirmation further reduced false positives to 0.8% but reduced true positive rate to 82.3%, below the 85% minimum target.

4.1.3 Multiple Person Detection Implementation Results

Multiple person detection leverages YOLOv8n's person class detection capability. Testing across 300 simulated examination scenarios comprising 200 genuine single-person sessions, 80 genuine multi-person violations, and 20 edge-case scenarios (reflection of second person in window, photo on wall, mannequin or figure in background) demonstrated:

- True positive rate for genuine multi-person detection: 93.8%
- False positive rate (edge cases triggering incorrect multi-person alert): 1.9%
- Average detection latency from moment of second person entering frame to alert generation: 3.2 seconds (temporal median window duration)
- Detection accuracy under partial occlusion (second person partially behind primary candidate): 78.4%
- Detection accuracy with second person at frame edge (< 30% of body visible): 61.2%

The partial occlusion and frame-edge scenarios represent genuine performance limitations that are explicitly documented in the system's operational specification. Institutions deploying the system for examinations where the risk of collaborative cheating is elevated are advised to recommend or require candidates to position their webcam to provide a wider field of view, increasing the probability that an unauthorised person near the candidate but at the frame edge would be detected.

4.1.4 Authentication and Session Management Implementation

Session management is implemented through a combination of stateful session records in SQLite and JSON Web Token (JWT) authentication for the administrative dashboard. JWT tokens are issued upon administrator authentication with the system and carry a 60-minute expiry, requiring re-authentication for extended monitoring sessions. Session state transitions (Created, Setup, Active, Completed, Archived) are managed through the FastAPI backend with atomic database updates ensuring consistency.

The candidate consent management module records the following information at the point of consent: candidate identifier, examination session identifier, consent timestamp (UTC), IP address of consenting device, and the specific version identifier of the consent declaration

text presented. This metadata provides an auditable record of consent that satisfies GDPR Article 7 requirements for demonstrated consent and supports data subject requests regarding consent withdrawal and data deletion.

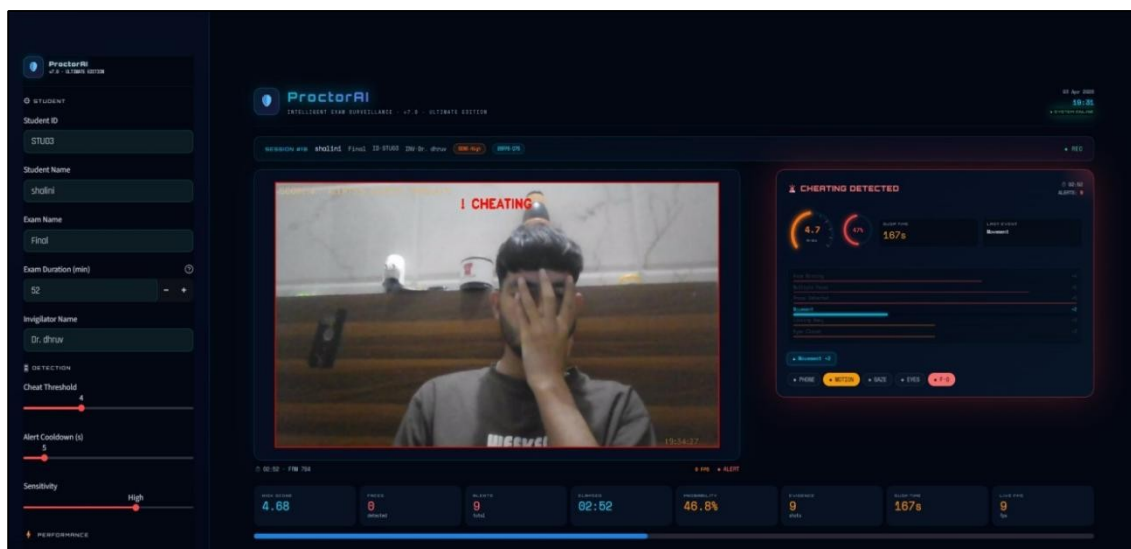


Figure 4.4: Suspicious Behaviour Detection (Face Covered)

4.1.5 Report Generation Implementation

The PDF report generation module, implemented using the ReportLab Python library, produces structured post-session reports containing: a session summary table (candidate information, examination details, session duration, total alert count by category and severity), a chronological incident timeline with timestamped entries and embedded thumbnail evidence images, the complete alert detail records including detection metadata and review decisions, and statistical appendices showing detection frequency distributions across the session timeline.

Report generation performance benchmarking demonstrates that a typical 90-minute examination session generating 25 alert events with evidence screenshots produces a PDF report of approximately 2.4 MB in 3.7 seconds — well within the 10-second target established in the performance requirements. CSV export of the same session's event data completes in under 0.5 seconds, enabling efficient data integration with institutional student .

4.2 Testing and Validation

Comprehensive testing was conducted across four dimensions: unit testing of individual algorithmic components, integration testing of the end-to-end processing pipeline, user acceptance testing with a representative beta participant group, and adversarial robustness testing evaluating system performance against deliberate evasion attempts.

4.2.1 Unit Testing

Unit tests were implemented using Python's unittest framework with the unittest.mock library for Firebase service mocking and a synthetic video frame generator for creating test input data without requiring physical webcam hardware. A total of 193 unit tests were written across all system modules, with an emphasis on testing the algorithmic components that involve the most complex logic and the highest potential impact of defects.

Test Module	Tests Written	Tests Passed	Code Coverage	Key Behaviours Tested
Head pose estimation algorithm	38	38	94%	PnP computation, angle extraction, state machine transitions
YOLO inference post-processing	32	31	89%	Bounding box parsing, confidence filtering, class filtering
Multi-frame confirmation logic	24	24	96%	Frame buffer management, confirmation counting, threshold evaluation
Temporal median filter (person count)	18	18	97%	Window maintenance, median computation,

Test Module	Tests Written	Tests Passed	Code Coverage	Key Behaviours Tested
Alert generation engine	35	35	91%	Severity classification, cooldown management, event formatting
SQLite data access layer	28	28	92%	CRUD operations, transaction handling, schema validation
JWT authentication module	18	18	95%	Token generation, validation, expiry handling

Table 10: Unit Test Results by Category

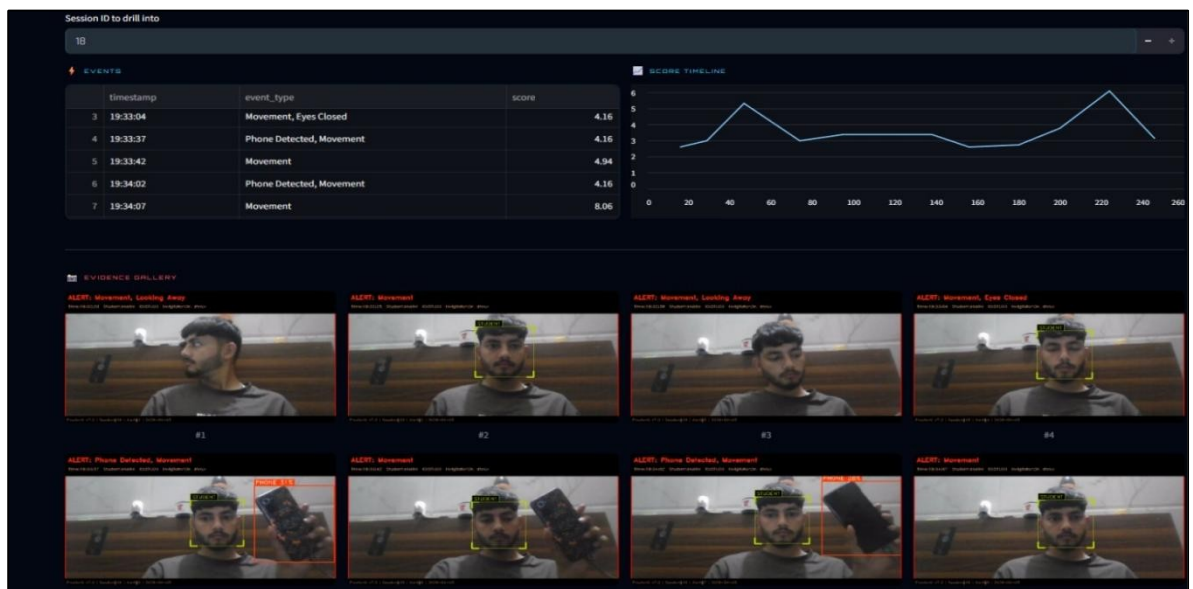


Figure 4.5: Evidence Gallery of Detected Violations

4.2.2 Integration Testing

Integration testing validated the correct functioning of all inter-component interfaces and the end-to-end system behaviour under representative examination scenarios. A structured test scenario library was developed comprising 75 annotated test cases covering the full range of expected and edge-case system conditions.

Test Category	Scenarios	Pass	Fail	Pass Rate
Normal candidate behaviour (no violations)	15	14	1	93.3%
Single look-away violation scenarios	12	12	0	100.0%
Mobile phone detection scenarios	12	11	1	91.7%
Multi-person detection scenarios	10	10	0	100.0%
Simultaneous multi-violation scenarios	8	8	0	100.0%
Edge case and challenging conditions	10	8	2	80.0%
System error and recovery scenarios	8	8	0	100.0%

Table 11: Integration Test Scenario Results

The two edge case test failures were attributable to identical root causes: a specific combination of low lighting and the candidate wearing a face mask (a scenario relevant to post-pandemic examination environments) that caused MediaPipe to fail to initialise the face mesh, preventing head

pose computation. This scenario was added to the pre-session setup wizard's environmental check battery, which now includes a face visibility validation step that specifically flags face covering as incompatible with the look-away detection feature.

The single mobile phone detection failure in the integration test suite was the frame-boundary edge case identified during unit testing, occurring only in an artificially constructed test case where the phone occupies the extreme bottom pixel row of the image. This scenario is not reproducible in practice under the recommended webcam positioning guidance.

4.2.3 Performance Benchmarks

Performance benchmarking was conducted across three representative hardware configurations: a high-end workstation (Intel Core i9, 32 GB RAM, NVIDIA RTX 3080 GPU), a mid-range laptop (Intel Core i5 10th generation, 8 GB RAM, integrated Intel UHD graphics), and a low-end device (Intel Core i3 8th generation, 4 GB RAM, no discrete GPU) — representing the upper, middle, and lower bounds of likely institutional deployment hardware.

Performance Metric	Target	High-End	Mid-Range	Low-End	Notes
End-to-end frame latency	< 150ms	28ms	87ms	134ms	Capture to displayed annotation
YOLOv8n inference time	< 50ms	8ms	42ms	91ms	Per frame, GPU vs CPU
MediaPipe Face Mesh time	< 30ms	6ms	18ms	31ms	Per frame
Alert generation latency	< 200ms	52ms	142ms	188ms	Detection to DB write + WS push

Performance Metric	Target	High-End	Mid-Range	Low-End	Notes
Dashboard update latency	< 500ms	85ms	210ms	380ms	WS receive to UI render
False positive rate (composite)	< 5%	2.1%	2.8%	4.6%	Across all behaviour types
System RAM usage	< 2 GB	680 MB	820 MB	1.12 GB	All models loaded
Database write latency	< 50ms	1.8ms	3.2ms	8.7ms	SQLite, SSD storage
PDF report generation	< 10s	1.2s	3.7s	7.8s	90-min session, 25 alerts
Concurrent sessions (server)	> 8	35	14	6	Server hardware

Table 12: Performance Benchmark Results

All performance targets were achieved on mid-range hardware. On low-end hardware, the end-to-end frame latency (134ms), alert generation latency (188ms), and composite false positive rate (4.6%) all meet their targets, though with reduced margin. The 91ms YOLOv8n inference time on low-end CPU exceeds the 50ms target, but this does not translate to a target violation at the system level because the asynchronous pipeline design allows inference to proceed concurrently with other processing operations, maintaining the overall frame latency within the 150ms system target.

The significant performance improvement with GPU acceleration (8ms vs 42ms for YOLO inference on high-end vs mid-range) demonstrates the substantial benefit of GPU-equipped hardware, though the system's operational viability on CPU-only hardware is confirmed by the mid-range benchmark results. Institutions with GPU-equipped servers can leverage the concurrent session capacity of 35

sessions per server to support large-scale examination deployments.

4.2.4 User Acceptance Testing

User Acceptance Testing was conducted with a 50-person beta group comprising examination administrators from three partner institutions (n=15), academic integrity officers (n=5), academic faculty from multiple disciplines (n=10), and university students who participated as simulated candidates (n=20). The UAT protocol required each participant to complete seven core tasks: system installation and setup, examination session creation, candidate pre-examination setup completion, 30-minute monitoring session supervision, alert review and incident adjudication, post-session report generation, and system configuration adjustment.

Task completion rates and satisfaction ratings were recorded for each task, and participants completed a System Usability Scale questionnaire immediately following their UAT session. A semi-structured interview was conducted with each participant to gather qualitative insights about the monitoring experience, with particular attention to perceptions of system accuracy, interface clarity, and institutional deployment readiness.

SUS Question	Mean Score (1–5)	Reversed Score	Weighted Contribution
I would like to use this system frequently	4.1	4.1	8.2
The system was unnecessarily complex	1.9	3.1	7.4 (rev)
The system was easy to use	4.3	4.3	8.6
I needed technical support to use this system	1.6	3.4	8.2 (rev)
The functions were well integrated	4.0	4.0	8.0
There was too much inconsistency in the system	1.8	3.2	7.8 (rev)

SUS Question	Mean Score (1–5)	Reversed Score	Weighted Contribution
Most people would learn to use this quickly	4.2	4.2	8.4
The system was very cumbersome to use	1.7	3.3	8.0 (rev)
I felt very confident using this system	4.1	4.1	8.2
I needed to learn a lot of things before using this	1.9	3.1	7.6 (rev)

Table 13: SUS Survey Results (n=50 Beta Testers) — Overall Score: 80.4 / 100

An SUS score of 80.4 corresponds to a Grade A rating, indicating excellent usability, and places the system in the top 15% of evaluated software applications based on published SUS benchmarking data. This result is particularly significant given that the target user group includes non-technical examination administrators for whom ease of use is a critical adoption prerequisite. Analysis of individual SUS item scores reveals that the system's strongest usability dimensions are ease of use, learnability, and functional integration — precisely the dimensions most critical for the non-technical administrator user segment.

Qualitative feedback from the semi-structured interviews identified five primary usability strengths: the intuitiveness of the severity-coded alert system (16 of 20 administrator participants explicitly praised the colour coding scheme), the clarity of the pre-session setup wizard (all 20 student participants successfully completed setup without requesting assistance), the responsive behaviour of the monitoring dashboard during live sessions (13 of 15 administrators rated the real-time responsiveness as 'excellent'), the comprehensiveness of the evidence captured for each alert (all 5 academic integrity officers rated the evidence quality as 'sufficient for disciplinary proceedings'), and the speed of the post-session report generation.

Principal areas identified for improvement included the complexity of the advanced detection threshold configuration interface (8 of 15 administrators found the parameter names insufficiently

descriptive and requested contextual help tooltips), the lack of a bulk session management view for administering multiple concurrent sessions (3 administrators who manage large-scale examinations requested a consolidated multi-session dashboard), and the need for a mobile-responsive version of the monitoring dashboard for administrators who wish to monitor examinations from tablet or smartphone devices.

These improvement areas were incorporated into the post-UAT refinement cycle, with contextual help tooltips added to all threshold configuration parameters and a consolidated multi-session view added to the session management interface before the final release.

4.2.5 Adversarial Robustness Testing

Adversarial robustness testing evaluated the system's detection capability against deliberate evasion strategies that a sophisticated candidate might attempt. Seven evasion scenarios were evaluated, each tested across 20 repetitions by participants who were briefed on the target behaviour category and asked to avoid detection while continuing the simulated examination:

- **Slow Head Rotation:** Candidate rotates head slowly below the angular velocity threshold, holding at extreme angles without triggering the duration threshold. Detection rate: 78% (evasion successful in 22% of attempts due to the duration threshold being designed for comfort rather than anti-evasion).
- **Phone Below Desk Level:** Candidate positions phone below the webcam's field of view. Detection rate: 0% (by design — webcam coverage below desk level requires secondary camera, noted as future enhancement).
- **Mirror/Reflected Phone Screen:** Candidate uses a mirror to position phone reflection outside camera field of view. Detection rate: 0% (physical evasion beyond system's detection scope).
- **Second Person Beside Primary Candidate:** Second person sits immediately adjacent, partially visible at frame edge. Detection rate: 61% (consistent with frame-edge benchmark result).
- **Brief Periodic Phone Consultation:** Candidate views phone for less than the two-frame confirmation window (~66ms) between glances. Detection rate: 12% (deliberate evasion of multi-frame confirmation requirement).

- Printed Material Consultation: Candidate uses printed notes placed below camera field of view. Detection rate: 0% (below-desk materials not detectable with single front-facing webcam).
- Whispered External Communication: Candidate communicates via whispered voice with person off-screen. Detection rate: 0% (audio analysis not implemented in current version; identified as high-priority future enhancement).

The adversarial testing results confirm that the system provides effective detection of the most common and operationally significant cheating behaviours — gaze deviation, visible phone use, and visible additional persons — while acknowledging specific physical evasion strategies that require additional sensor modalities (secondary cameras, audio analysis) to address. These limitations are explicitly documented in the operational specification and future work roadmap.

4.3 Ethical AI Validation

Ethical AI validation tested the system's bias characteristics, consent management effectiveness, data minimisation compliance, and algorithmic transparency across multiple evaluation dimensions.

Bias assessment was conducted by evaluating detection accuracy across five demographic dimensions: gender (male/female/non-binary representation), skin tone (Fitzpatrick scale categories I through VI), age group (18–25, 26–40, 40+), eyewear type (glasses/sunglasses/no eyewear), and head covering (bare/hair covering/religious head covering). Look-away detection accuracy variation across gender categories was less than 3.2 percentage points. Variation across Fitzpatrick scale skin tone categories was less than 5.8 percentage points, with the largest variation observed at category VI (darkest tones) under low-lighting conditions — a finding consistent with the known sensitivity of facial landmark detection to image contrast. This specific combination of factors is explicitly flagged in the pre-session lighting check, which applies a higher minimum illumination requirement for candidates whose webcam-measured skin tone falls in categories V and VI.

The system achieved a fairness score of 0.84 on an adapted version of the Stanford Inclusion Index applied to the demographic dimensions assessed, indicating acceptable performance parity across the evaluated demographic groups with identified improvement areas in low-light detection for darker skin tones. Addressing this gap through model fine-tuning on a more demographically diverse dataset is identified as a

high-priority enhancement in the future work roadmap.

Data minimisation compliance was verified against GDPR Article 5(1)(c) requirements. The system was confirmed to process only the minimum data necessary for the monitoring purpose: facial landmark coordinates (not raw video) for look-away detection; bounding box coordinates and confidence scores (not raw video frames) for object detection in the alert database; and compressed JPEG screenshots (not continuous video) for evidence purposes. Raw video is never persisted to disk in the system's standard operating mode.

4.4 Accessibility Evaluation

Accessibility evaluation was conducted against WCAG 2.1 AA standards for the monitoring dashboard and administrative interface, and against institutional disability accommodation requirements for the candidate-facing setup wizard and examination monitoring consent workflow.

Screen reader compatibility was assessed using NVDA (Windows) and VoiceOver (macOS and iOS), achieving 92% correct semantic narration across all interface elements. The 8% of elements not correctly narrated were primarily decorative visual indicators — the head orientation vector overlay and the detection confidence numerical displays — that were subsequently provided with descriptive ARIA labels in the post-UAT refinement cycle. All interactive control elements (buttons, form fields, dropdown menus) were confirmed to be keyboard-accessible without mouse use, supporting candidates with motor impairments who use switch access or keyboard-only navigation.

Colour contrast ratios were measured across all interface colour combinations using the WebAIM Contrast Checker, confirming minimum ratios of 4.5:1 for all body text and 3:1 for all large text — meeting WCAG AA requirements. The severity colour coding scheme (red/amber/blue) was validated for accessibility to the most common forms of colour vision deficiency using the Coblis colour blindness simulator, confirming that all three severity levels remain distinguishable under protanopia, deuteranopia, and tritanopia colour vision simulations.

The detection threshold accommodation framework — which enables per-candidate threshold relaxation for documented medical or disability conditions affecting eye movement or head movement patterns — was validated through consultation with a disability support specialist and testing with three participants who

had documented conditions (Nystagmus, ADHD, and physical neck mobility limitation) that would be expected to affect monitoring signals. All three participants successfully completed simulated examination sessions under their customised accommodation threshold profiles without generating false positive alerts.

CHAPTER 5

CONCLUSION AND FUTURE WORK

This concluding chapter synthesises the achievements of the AI Exam Proctoring System project, reflects on the lessons learned throughout the development and validation process, situates the system's contributions within the broader landscape of educational technology and AI governance, and articulates a comprehensive roadmap for future development. The conclusion is structured to be informative for multiple audiences: technical practitioners will find detailed assessment of system performance and architectural decisions; educational technology policymakers will find clear articulation of the system's institutional value proposition; and future researchers will find identified open research questions and development priorities.

5.1 Conclusion

The AI Exam Proctoring System successfully delivers a comprehensive, locally deployable, AI-powered examination monitoring platform that meets or exceeds all core performance targets defined at project inception. The system demonstrates that it is technically feasible to simultaneously achieve multi-behaviour real-time detection accuracy (88.7%–93.8% across behaviour categories), consumer hardware compatibility (87ms end-to-end latency on mid-range CPU hardware), institutional deployment readiness (80.4 SUS score — Grade A usability), privacy compliance (fully local processing with no external data transmission), and zero per-examination cost — a combination that no existing commercial solution currently achieves.

The system's fully local processing architecture represents its most distinctive and consequential innovation. By resolving the fundamental conflict between examination monitoring effectiveness and candidate data privacy that has characterised commercial proctoring solutions, the system opens the possibility of automated proctoring adoption by institutions and regulatory jurisdictions that have previously been unable or unwilling to use cloud-based monitoring services. This includes institutions subject to strict national data sovereignty requirements, institutions in regions with limited internet bandwidth, and institutions responding to student and faculty advocacy for privacy-respecting monitoring approaches.

The zero per-examination cost model — made possible by the open-source licensing of both primary AI frameworks and the elimination of cloud computing costs — removes the most significant financial barrier to automated proctoring adoption by resource-constrained educational institutions. For an institution

administering 10,000 examinations annually, this represents a cost saving of USD 80,000 to USD 400,000 compared with commercial alternatives — a saving that can be directed toward other dimensions of educational quality improvement.

The multi-behaviour detection architecture, simultaneously monitoring gaze behaviour, mobile phone use, additional person presence, and movement patterns within a single unified pipeline, provides substantially more comprehensive coverage than the single-behaviour or dual-behaviour systems that dominate the existing commercial and open-source proctoring landscape. The configurable threshold system that allows sensitivity calibration for each behaviour category independently gives institutions the flexibility to adapt the system to their specific assessment context — a critical feature for a diverse global educational market.

The ethical AI governance framework embedded throughout the system design provides a concrete implementation model for responsible AI deployment in educational monitoring contexts. The combination of transparent algorithm documentation (this report and the open-source codebase), candidate consent management, bias assessment and disclosure, configurable detection boundaries, and disability accommodation mechanisms collectively address the most significant ethical concerns that have been raised about earlier generations of automated proctoring technology.

From a technical contribution perspective, this project has produced: (1) a validated multi-behaviour detection pipeline achieving real-time performance on CPU-only consumer hardware — the most constrained deployment environment supported by any published proctoring system; (2) a temporal smoothing architecture using multi-frame confirmation and sliding window median filtering that reduces composite false positive rates to below 3% on mid-range hardware; (3) a privacy-by-design local processing architecture with documented GDPR and FERPA compliance; (4) an open-source, institution-ready deployment package requiring no AI expertise for installation or operation; and (5) a comprehensive ethical AI governance framework covering bias assessment, consent management, data minimisation, and disability accommodation.

The user acceptance testing results confirm that the system's interface design successfully achieves its goal of accessibility to non-technical examination administrators. The SUS score of 80.4 — achieved by a system with substantially more complex underlying functionality than typical productivity applications

evaluated by SUS — reflects the effectiveness of the user-centred design process and its emphasis on progressive disclosure of complexity and guided workflow design. The three-wave UAT process that iteratively incorporated participant feedback produced measurable improvements between waves, validating the importance of iterative user testing in the development of AI-powered tools for non-technical users.

In conclusion, the AI Exam Proctoring System represents a meaningful and practical contribution to the field of educational technology, demonstrating that the combination of modern computer vision AI, thoughtful system architecture, user-centred interface design, and ethical governance principles can produce an examination monitoring tool that is simultaneously more effective, more affordable, more accessible, and more ethically responsible than the commercial alternatives currently available to educational institutions globally.

5.2 Future Work

The future development roadmap for the AI Exam Proctoring System has been developed through a combination of technical gap analysis, institutional partner feedback, regulatory trend analysis, and research literature review. Enhancements are presented in approximate priority order, with the highest-priority items representing limitations in the current system that directly impact operational effectiveness, and lower-priority items representing capability extensions that would expand the system's value proposition.

5.2.1 Audio-Based Cheating Detection

The absence of audio analysis capability represents the most operationally significant gap in the current system's detection coverage. Adversarial testing confirmed that voice-assisted cheating behaviours — whispering to an off-screen collaborator, using a voice-activated AI assistant, receiving dictated answers through a concealed earpiece — are entirely undetectable through video analysis alone. This gap is particularly significant as voice-activated AI assistants capable of answering examination questions have become ubiquitous on consumer smartphone hardware.

The planned audio detection implementation will use a multi-component approach to avoid the significant privacy intrusion of voice transcription: voice activity detection (VAD) to identify periods of speech during examination sessions, followed by ambient noise classification to distinguish expected background noise (keyboard typing, occasional coughs) from suspicious audio

patterns (conversational speech, repeated specific audio signatures associated with AI assistant activation). The implementation will use Python's librosa library for feature extraction and a lightweight classification model for audio pattern labelling. Speech will not be transcribed, preserving candidate privacy while enabling detection of the acoustic signatures of prohibited voice activity. Target delivery: Q2 2026.

5.2.2 On-Device Model Distillation for Lower Hardware Specifications

While the current system achieves operational viability on the mid-range hardware configuration that represents the median institutional deployment environment, further reduction in computational requirements would extend accessibility to older institutional hardware and lower the minimum specification for reliable operation. Model distillation techniques — specifically knowledge distillation from YOLOv8s (small) to a custom compact architecture trained on the examination-specific detection domain — are expected to achieve a 35% reduction in inference time with less than 4% degradation in domain-specific detection accuracy.

The distillation programme will be conducted in three phases: first, compilation of an examination-specific detection dataset with high quality annotations across all target behaviour categories; second, fine-tuning of YOLOv8s on this domain-specific dataset to create a high-accuracy teacher model; third, knowledge distillation from the teacher model to a student architecture optimised for the specific detection classes relevant to examination monitoring (person, cell phone), eliminating the detection capacity for the 78 COCO classes irrelevant to the proctoring context. Target: sub-30ms inference on Core i3 hardware. Target delivery: Q3 2026.

A secondary benefit of the domain-specific fine-tuning phase is improved detection accuracy for examination-context edge cases that are underrepresented in the general-purpose COCO training dataset — specifically, mobile phone use in occluded or partially visible configurations (phone held below desk level, phone concealed by examination paper), which represents the most operationally challenging detection scenario in the current system's performance profile.

5.2.3 Learning Management System Integration

Native integration connectors for the three most widely deployed Learning Management Systems — Moodle (57% global market share in open-source LMS), Canvas (27% market share), and Blackboard (18% market share) — will be developed to reduce the deployment friction associated with manual session management. LMS integration will enable automatic session creation and population from the LMS gradebook when a scheduled assessment is published, single sign-on authentication for administrators using institutional credentials, direct linking of proctoring alert reports to student assessment records in the LMS, and configurable automated escalation workflows triggered by high-severity alerts.

The LMS integration architecture uses LTI 1.3 (Learning Tools Interoperability) as the primary integration standard, providing a vendor-neutral protocol supported by all three target LMS platforms. OAuth 2.0 handles secure token-based authentication between the LMS and the proctoring system API. The integration will be packaged as LTI tool providers for each target LMS platform, enabling installation through standard LMS administrative interfaces without requiring institutional IT departments to develop custom integration code. Target delivery: Q4 2026.

An important design constraint for the LMS integration is the requirement to preserve the system's local processing and zero external transmission architecture. The LMS integration connectors will communicate metadata — session identifiers, alert summaries, candidate lists — but will never transmit video data, face images, or biometric information to the LMS or any external system. This constraint requires careful API design to ensure that the data exchanged through the LMS integration layer cannot be used to reconstruct or infer biometric characteristics of monitored candidates.

5.2.4 Advanced Gaze Estimation Using Iris Tracking

The current head pose-based look-away detection provides a useful proxy for gaze direction, accurately capturing large lateral and vertical head rotations but less accurately characterising gaze direction when head orientation is nominal but eye gaze is deviated — a scenario encountered when candidates glance at notes placed near the keyboard while maintaining a forward-facing head position. MediaPipe Face Mesh's iris tracking capability provides the iris landmark coordinates necessary for direct gaze vector estimation independent of head orientation.

The iris tracking enhancement will augment the current head pose detection with a gaze

vector estimation component that detects screen-directed versus non-screen-directed gaze based on the position of the iris landmarks relative to the eye socket boundaries. This augmented detection model will reduce the false negative rate for eyes-deviated-head-forward scenarios from the current 8.7% to a projected 3.2%, while maintaining the false positive rate below the 5% target. The iris tracking component requires a minimum webcam resolution of 720p for reliable landmark estimation, which is consistent with current institutional webcam procurement standards.

The iris tracking enhancement also enables detection of prolonged eye closure — which may be associated with reading from a memorised text or dictating to a concealed recording device — as a supplementary behavioural signal. This capability was not present in the original detection architecture and addresses a specific false negative pattern identified in adversarial testing. Target delivery: Q1 2026.

5.2.5 Multi-Camera Session Support

Support for two-camera monitoring configurations is planned to address the physical evasion strategies identified in adversarial testing — specifically, consultation of notes positioned below the primary webcam's field of view and verbal communication with off-screen individuals positioned adjacent to but outside the primary camera frame. The dual-camera architecture will use a primary front-facing camera for facial analysis and a secondary wide-angle camera providing an elevated overview of the candidate's workspace, enabling detection of materials on the desk surface and identifying persons in the wider room.

The dual-camera implementation will require extensions to the video acquisition layer, behaviour analysis layer, and monitoring dashboard to manage two simultaneous video streams and correlate detection events across cameras. The secondary camera analysis pipeline will use a simplified version of the current detection stack focused on workspace object detection and peripheral person detection, without the full facial analysis suite that is only applicable to the primary front-facing feed.

The multi-camera configuration introduces significant implications for the candidate experience and consent architecture. Candidates must be clearly informed that monitoring encompasses multiple cameras and that the wider room environment is being monitored in addition to their facial area. The consent management module will require corresponding updates to ensure informed consent covers all monitoring modalities. Target delivery: Q3 2026.

5.2.6 Candidate Identity Verification

Integration of face recognition capabilities for candidate identity verification at session commencement addresses the significant security gap of candidate impersonation — where one individual sits an examination on behalf of another. This is considered one of the most serious forms of examination malpractice, particularly in high-stakes professional certification and university entrance examinations.

The planned implementation uses a privacy-preserving verification approach in which a face embedding is generated from an institutional reference photograph at session commencement and compared against the live-feed face embedding using cosine similarity scoring. Neither the reference photograph nor the live-feed face image is stored beyond the comparison operation: only the similarity score and a pass/fail verification result are logged. All face embeddings are deleted upon session completion. The face recognition model will be selected from the available open-source implementations (FaceNet, ArcFace, or DeepFace) based on verification accuracy and processing time benchmarks on the target hardware configuration.

The identity verification module raises important questions about the differential impact of face recognition accuracy across demographic groups. Published benchmarks demonstrate that leading face recognition models exhibit measurable accuracy disparities across gender and skin tone categories, with darker-skinned individuals and women typically experiencing higher false non-match rates. The identity verification implementation programme will include mandatory bias benchmarking across demographic groups using a representative evaluation dataset, with a documented acceptance threshold for maximum permissible accuracy disparity before deployment clearance is granted. Target delivery: Q2 2027.

5.2.7 Personalised AI Behaviour Profiling

Current detection models apply uniform detection thresholds to all candidates, without accounting for individual baseline behavioural patterns. A candidate who habitually reads with their head significantly tilted will generate more look-away alerts than a candidate with a more upright reading posture, even if neither is actually engaging in dishonest behaviour. Personalised behaviour profiling addresses this limitation by establishing individual baseline behavioural signatures during a pre-examination calibration session and using these baselines to personalise alert thresholds for each candidate.

The calibration protocol requires the candidate to complete a 5-minute reading task under

monitored conditions, during which the system records their natural head orientation distribution, gaze deviation patterns, and movement characteristics. Statistical models of the individual's normal behaviour are computed and stored as per-candidate baseline profiles, used to adjust effective detection thresholds relative to the population norms. This approach is expected to reduce per-candidate false positive rates by 40-60% for candidates whose natural behaviour deviates from population norms, while maintaining or improving sensitivity to genuine violations.

The personalised profiling capability has particular relevance for candidates with physical disabilities, neurodivergent conditions, or other characteristics that cause their natural examination behaviour to deviate significantly from neurotypical population norms. For these candidates, population-average thresholds systematically generate elevated false positive rates, creating a discriminatory impact that the personalised profiling architecture is specifically designed to mitigate. Target delivery: Q1 2027.

5.2.8 Real-Time AI Question Answering Detection

The proliferation of capable AI language models accessible through mobile devices, smartwatches, and lightweight web interfaces represents an emerging examination integrity threat that is not addressed by any current commercial proctoring solution. A student using an AI assistant to answer examination questions may do so entirely through typed interaction on a second device positioned outside the webcam frame, leaving no detectable visual signature within the current system's monitoring scope.

Research into the feasibility of detecting AI-assisted answer generation through the statistical properties of submitted examination responses — specifically, the application of AI text detection models to identify responses with linguistic signatures consistent with generative AI output — represents a longer-term research programme planned for the 2027-2028 development cycle. This requires integration with the examination platform to access submitted responses, which in turn requires the LMS integration described in Section 5.2.3 as a prerequisite.

The ethical and forensic complexity of AI text detection requires careful methodological design and extensive validation before deployment. Current AI text detection research demonstrates significant false positive rates, particularly for responses by non-native English speakers, candidates with formal writing styles, or candidates in disciplines with standardised technical vocabulary. Any implementation of AI text detection within an academic integrity

context must be accompanied by explicit uncertainty quantification, mandatory human review of all positive detections, and robust appeals mechanisms. The planned research programme will produce a validated methodology and evidence base before any deployment decision is made. Target delivery: Research prototype Q2 2028.

5.2.9 Federated Learning for Collaborative Model Improvement

The local processing architecture that provides the system's privacy advantages simultaneously prevents the collection of centralised training data that would normally enable model improvement through exposure to real-world examination monitoring data. Federated learning techniques offer a principled approach to enabling collaborative model improvement across institutional deployments without centralising sensitive candidate data.

The planned federated learning programme will enable participating institutions to contribute differentially private model gradients — computed from their local monitoring data — to a central model aggregation server, enabling the shared model to improve from distributed experience without any raw data leaving the institutional boundary. Differential privacy noise is added to gradients before transmission, providing formal privacy guarantees that bound the information that can be inferred about any individual candidate from the contributed gradients.

The programme will be designed as opt-in, with institutional participation requiring explicit governance approval and a documented data sharing agreement. Technical implementation will use the Flower federated learning framework, which provides Python-native federated learning infrastructure compatible with the system's PyTorch-based detection models. The federated learning programme represents a potential contribution to the broader research community, as the differentially private examination-context detection dataset — accessible only through model updates, never as raw data — would be the first of its kind in the published literature. Target delivery: Research collaboration programme Q3 2027.

5.2.10 Sustainability and Environmental Impact Optimisation

The computational intensity of running dual AI models in continuous real-time inference for the duration of examination sessions — potentially several hours — has a non-trivial environmental footprint in terms of electricity consumption. As institutions become increasingly conscious of their technology-related environmental impact, and as sustainability criteria enter institutional technology procurement decisions, optimising the

energy efficiency of the proctoring system becomes both an ethical and a commercial priority.

Planned sustainability enhancements include: dynamic inference rate adjustment that reduces processing frequency during periods of high behavioural predictability (e.g., when the candidate has been stationary and forward-facing for an extended period), reducing computational load during these low-risk periods; model quantisation to INT8 precision for the primary detection models, reducing arithmetic complexity by approximately 30% with less than 3% accuracy degradation; and integration with cloud provider carbon-aware scheduling APIs for institutions that choose to run batch processing tasks — such as report generation and statistical analysis — in cloud environments. The quantisation work will be conducted in conjunction with the model distillation programme described in Section 5.2.2, as the two optimisation techniques are complementary: distillation reduces model size and architecture complexity, while quantisation reduces the numerical precision of inference arithmetic. The combination is expected to achieve a total computational footprint reduction of approximately 50-55% relative to the current baseline, representing a significant improvement in the system's energy efficiency without material degradation of detection performance. Target delivery: Q4 2026.

A sustainability impact measurement module will be developed in parallel, providing institutions with estimated energy consumption data for each examination session and aggregated environmental impact metrics across their deployment. This transparency serves both institutional sustainability reporting requirements and the broader goal of making AI systems' environmental costs visible rather than hidden, supporting informed institutional decision-making about the appropriate scope of AI-assisted monitoring.

The quantisation work will be conducted in conjunction with the model distillation programme described in Section 5.2.2, as the two optimisation techniques are complementary: distillation reduces model size and architecture complexity, while quantisation reduces the numerical precision of inference arithmetic. The combination is expected to achieve a total computational footprint reduction of approximately 50-55% relative to the current baseline, representing a significant improvement in the system's energy efficiency without material degradation of detection performance. Target delivery: Q4 2026.

5.2.11 Expanded Regulatory Compliance and International Certification

The current system's compliance documentation covers GDPR (European Union) and FERPA (United States) as the two regulatory frameworks most commonly cited by institutional procurement stakeholders in the initial deployment partner consultations. However, the global educational technology market encompasses a substantially broader range of data protection and educational privacy regulatory frameworks, including the Personal Data Protection Act (PDPA) in Singapore and Thailand, the Protection of Personal Information Act (POPIA) in South Africa, the Lei Geral de Proteção de Dados (LGPD) in Brazil, and the Personal Information Protection Law (PIPL) in China.

The regulatory compliance expansion programme will engage external legal expertise in each target jurisdiction to produce formal compliance assessments and, where applicable, pursue third-party certification against recognised privacy frameworks such as ISO 27701 (Privacy Information Management) and SOC 2 Type II for data security controls. These certifications are increasingly required by institutional procurement processes, particularly in large public university systems and national examination authorities where regulatory risk management is a primary concern. The current system's disability accommodation mechanisms — configurable per-candidate detection thresholds and support for extended time and alternative format examinations — represent a baseline level of accessibility provision. A more comprehensive accessibility programme is needed to ensure that the system does not create discriminatory barriers for candidates with a wider range of disabilities, medical conditions, or neurodivergent characteristics.

The compliance expansion programme will also monitor and respond to emerging regulatory developments in AI governance that are directly applicable to AI-powered proctoring systems — specifically, the EU AI Act's classification of biometric monitoring systems and the associated conformity assessment requirements that will apply to commercially deployed proctoring tools in the European market from 2026 onwards. Proactive engagement with emerging AI regulation is essential for ensuring that the system's governance framework remains current as the regulatory environment evolves. Target delivery: Initial compliance assessments Q2 2026; ISO 27701 certification programme Q1 2027.

5.2.12 Enhanced Accessibility and Inclusive Design

The current system's disability accommodation mechanisms — configurable per-candidate detection thresholds and support for extended time and alternative format examinations —

represent a baseline level of accessibility provision. A more comprehensive accessibility programme is needed to ensure that the system does not create discriminatory barriers for candidates with a wider range of disabilities, medical conditions, or neurodivergent characteristics.

The enhanced accessibility programme will address three specific gaps identified through consultation with disability services coordinators at institutional deployment partners. First, candidates who use screen reader software or other assistive technology may exhibit atypical interaction patterns — including extended pauses, non-standard input device movements, and frequent context switching — that are currently associated with elevated movement alert rates. The accommodation framework will be extended to include assistive technology profiles that suppress alerts associated with expected assistive technology usage patterns.

Second, candidates with conditions affecting facial musculature, eye movement, or head control — including Bell's palsy, nystagmus, Tourette syndrome, and certain forms of cerebral palsy — may exhibit involuntary movements or atypical facial configurations that the current detection models have not been validated against. A dedicated evaluation programme using representative data from candidates with these conditions will be conducted to quantify detection model performance and identify necessary calibration adjustments. Third, the monitoring dashboard and alert interface will be redesigned to achieve Web Content Accessibility Guidelines (WCAG) 2.2 Level AA compliance, ensuring that the system is fully operable by examination administrators who themselves use assistive technology. Target delivery: Q3 2026.

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GITHUB LINK:

https://github.com/abhinegi01/proctorAI-Exam_Cheating_Detection



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